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Al-razi University
Faculty of Computer Science and IT

الجمهورية اليمنية
وزارة التعليم العالي والبحث العلمي
جامعة الرازي
كلية الحاسوب وتقنية المعلومات
قسم تقنية المعلومات

Water Services System for automating processes between consumers, water providers, and wells owners

Submitted by:

1. Mohammed Tawfik Al-Albi (2021010285)
2. Nasser Nedhal Wahan (2020011411)
3. Abdul-Aziz Yahya Al-Wared (2020011409)

Supervisor
Assoc Prof. Yahya Al-Ashmouri

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Dedication

This project is dedicated to our families and friends, whose unwavering support and encouragement have served as our guiding light throughout this journey. To our cherished parents, we extend our heartfelt gratitude for instilling in us the values of perseverance and hard work. Your belief in our potential has consistently motivated us to strive for excellence in all our endeavors.

We also dedicate this work to the communities we aspire to serve, particularly those facing significant challenges in accessing essential water services. Your resilience inspires us, and we hope our efforts contribute to meaningful improvements in your lives.

May this project stand as a testament to the collective support we have received and our commitment to making a positive impact. Together, we aim to foster a brighter future for all, guided by compassion and a desire for equitable access to vital resources.

Acknowledgment

We would like to express our deepest and most sincere gratitude to Assoc. Prof. Yahya Al-Ashmouri for his exceptional guidance, unwavering support, and invaluable insights throughout the duration of this project. His expertise not only shaped our work but also inspired us to achieve higher standards of excellence.

We extend our heartfelt appreciation to our families, whose unwavering encouragement and belief in our abilities have been a constant source of motivation. Their support has provided us with the strength to overcome challenges and pursue our goals with determination.

A special thanks is also due to our colleagues, whose collaboration and stimulating discussions have enriched our research experience. The diverse perspectives and shared knowledge have significantly enhanced the depth of our work.

Furthermore, we acknowledge the invaluable resources and tools that facilitated our development process. These resources have played a crucial role in enabling us to conduct thorough research and implement effective solutions.

Each contribution, whether large or small, has played a vital role in bringing this project to fruition. We are genuinely grateful for the opportunity to learn and grow through this experience, and we look forward to applying these lessons in our future endeavors.

Thank you to everyone who supported us along the way; your encouragement has made a profound impact on our journey.

Examiner Committee

We are pleased to present our project to the esteemed Examiner Committee, comprised of distinguished professionals who bring a wealth of knowledge and experience to our evaluation process. The committee members are:

- Dr. Yahya Al-Ashmouri
- Dr. Khaled Al-Wasabi
- Dr. Iman Al-Shaari

We are honored to have such an esteemed committee review our project, and we look forward to their constructive feedback and insights, which will undoubtedly aid us in our pursuit of academic and professional excellence.

Signature	Signature	Signature
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Abstract

This project entails the development of an integrated application comprising four primary interfaces: **User Interface**, **Consumer Interface**, **Well Owner Interface**, and **Administrator Interface**. The application is designed to support and manage a water distribution service, facilitating seamless communication and interaction among consumers, water providers, and well owners.

The system aims to enhance operational efficiency and transparency by enabling consumers to search for available water providers or wells, access comprehensive reports detailing water consumption, unit costs, and provider-specific transactions. Additionally, consumers receive real-time notifications regarding water usage and the associated charges through the application.

Water providers benefit from the platform through streamlined documentation of water distribution activities, including transactions between consumers and wells. The application assists providers in managing their operations more effectively and ensuring accurate service delivery.

Well owners are integrated into the system through functionality that allows them to record water meter readings directly within the application. These readings are then used to generate detailed usage reports for consumers, thereby supporting timely and accurate billing.

The Administrator Interface serves as a central control panel for managing the application, overseeing system operations, managing users, and ensuring data consistency and system integrity.

By consolidating these functions into a single application, the system improves service coordination, enhances transparency, and supports the effective management of water resources.

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FCIT	Faculty of Computer and Information Technology
IT	Information Technology
SDLC	Software Development Life Cycle
UML	Unified Modeling Language
OTP	One Time Password
GPS	Global Positioning System
API	Application Programming Interface
JSON	JavaScript Object Notation
AOT	Ahead-of-Time Compilation
HCI	Human-Computer Interaction

1Table 1 list of Abbrevaitions

1. Introduction

1.1 Introduction

With the new development of technology, everything around us started to change. So, people cannot cope with those changes if they keep using the traditional and old methods from the old generation. Nowadays, technology has intertwined with the way of dealing with tasks in our lives, such as ordering food, buying things, ways of payments, and more.

It becomes necessary for all businesses to make it easy for users/ consumers as it can facilitate all the services they need in their daily life.

one of the things that make the consumers struggle is getting water service from the water provider or well owner. not only the consumer who face difficulties, but the well owner also cannot track the units consumed unless he goes to the consumer and registers the units the consumer used, then matches it with the previous unit reading. This operation takes time and effort and sometimes gets into mistakes. So, it is necessary to automate the whole operation to save time and effort.

1.2 Project definition

The proposed project is a specialized application designed to enhance the management and delivery of water services by automating communication and operational workflows among consumers, water providers, and well owners. The system encompasses a range of functionalities, including the ability to locate nearby wells, generate detailed water consumption reports, facilitate online payments, and streamline essential tasks such as meter readings and order processing. To support these functions, the application incorporates technologies such as GPS for location tracking, push notifications for real-time updates, and barcode scanning for accurate and efficient data entry. The primary objective of the system is to optimize time management, minimize human error, and improve overall operational efficiency for all stakeholders involved in the water supply chain.

1.3 Problem Statement

The primary challenge addressed by this project lies in the inefficiencies and communication gaps experienced among consumers, water providers, and well owners in the current water delivery process. These issues hinder smooth operations and result in time-consuming and error-prone practices. The specific problems can be summarized as follows:

1. **Difficulty in Locating Water Sources**: Consumers often face challenges in identifying and accessing the nearest available water wells due to the lack of centralized or location-based search tools.

2. **Lack of Transparency in Consumption and Availability**: Both consumers and water providers experience difficulties in tracking the amount of water consumed over specific periods. Additionally, there is a lack of real-time information regarding the availability of water, which complicates planning and usage.

3. **Manual Payment Processes**: The current reliance on manual fee collection requires significant time and effort from both consumers and providers, leading to inefficiencies and potential inaccuracies in financial transactions.

4. **Time-Consuming Meter Reading and Calculation**: Well owners and water providers spend considerable time and effort reading water meters and calculating consumption manually, which can result in delays and data entry errors.

These challenges underscore the need for an automated, integrated system that facilitates seamless communication, accurate data tracking, and efficient service delivery among all stakeholders involved in the water distribution process.

1.4 Project Objectives

The primary objective of this project is to automate and streamline the processes of communication and service delivery among consumers, well owners, and water providers. In response to the previously identified challenges, the system aims to achieve the following:

1. **Enable Location-Based Search**: Utilize GPS technology within the consumer interface to display the nearest available water wells based on the user's current location, thereby improving accessibility and decision-making.
2. **Provide Comprehensive Reporting and Notifications**: Facilitate the generation of detailed reports for both consumers and water providers, including information on water consumption units, usage duration, and provider-specific transactions. The system will also support the delivery of real-time notifications to consumers regarding water availability and consumption updates.
3. **Implement Online Payment Services**: Integrate secure online payment functionality to replace manual fee collection, thereby reducing effort and increasing the efficiency and accuracy of financial transactions.
4. **Integrate Barcode Functionality for Accurate Data Entry**: Generate unique barcodes for water meters and provider vehicles to be scanned by well owners. This will allow for quick and accurate recording of water consumption units directly into the system, minimizing errors and manual data handling.

Through these objectives, the project seeks to enhance operational efficiency, reduce human error, and foster effective communication between all parties involved in the water delivery ecosystem.

1.5 Project Important

The implementation of this project is crucial for improving the efficiency and reliability of water delivery services by streamlining communication and operational processes among consumers, water providers, and well owners. The proposed system offers a centralized platform that simplifies interactions and enhances transparency across all stakeholders.

For consumers, the application provides a convenient means to communicate with water providers and well owners, significantly reducing the time and effort involved in securing water services. Additionally, consumers will benefit from access to detailed operational reports, including consumption data and transaction history, which enhances awareness and accountability.

Water providers will be able to manage relationships with multiple wells more effectively and receive daily reports detailing water units consumed. This supports better planning, improved service delivery, and efficient resource allocation.

Well owners, in turn, will utilize the application to monitor and manage reports from both consumers and providers. The system also simplifies the process of recording water consumption, offering an accurate and time-efficient alternative to manual methods.

Furthermore, the inclusion of a built-in accounting manual within the application serves as a reference point for users, promoting financial clarity and consistency. This feature contributes to reducing the risk of fraud and minimizing the chances of financial mismanagement, thereby fostering trust and accountability within the system.

1.6 Project Scope and Limitations

The study focuses on automating the processes between consumers, water providers, and wells owners. In addition, ease the process of reading and calculating the consumed units for consumers, water providers, and wells owners.

There were some difficulties that fasces during the developing and implementation of the application, one of the challenges that faced us was finding the google API to connect to google maps, which we have already solved. Another difficulty that we encountered was implementing the payment API into our app for the online payment that which the exchange companies didn’t accept to connect to our app until the development process is completed and the app is ready to work. And they want a commercial guarantee for the app.

1.7 Tools used in the project

The tools used un the project :

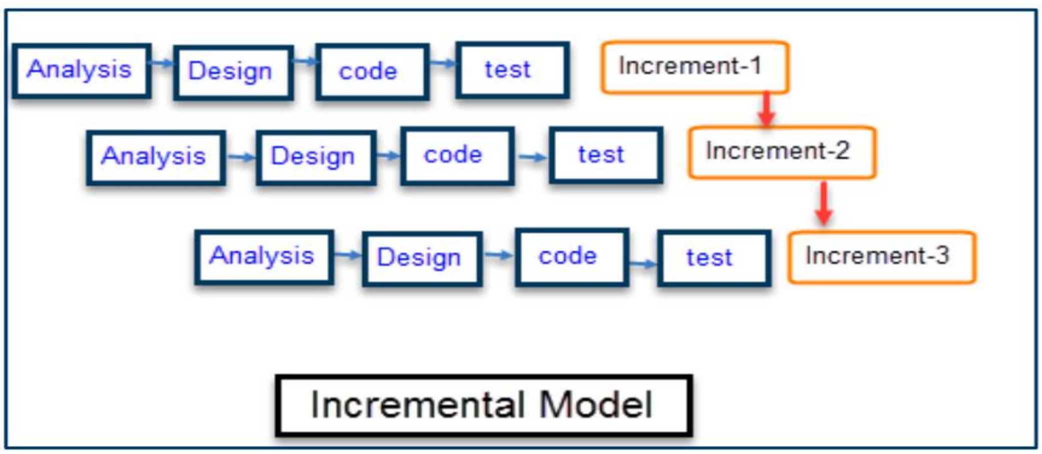
Name	Usage
Windows 10 or 11	To run the needed environments and compliers.
Android studio	To code and compile
Visual studio code	To code and compile
Flutter SDK 3.34	The flutter platform for coding
Microsoft office 2019	For documentation
ServerXampp	For serves
My Sql	database
Canva	Design
Book	Write
modem	For connect to internet
laptop	To do all thing
room	for meeting

2 Table 1-1 (tools)

1.8 Research Methodology

The methodology used is the incremental model (Model Incremental). Incremental Model is a process of software development where requirements divided into multiple standalone modules of the software development cycle as shown in Figure (1-1). In this model, each module goes through the requirements, design, implementation and testing phases. Every subsequent release of the module adds function to the previous release. The process continues until the complete system achieved.

- Reasons for choosing the methodology:
  1. Clarity and full understanding of the requirements for the system.
  2. The ability of the system to develop.
  3. Flexibility of the methodology in keeping pace with technology, and its scalability for development.



1Figure 1-1 Incremental Model

1.9 Timetable for the project

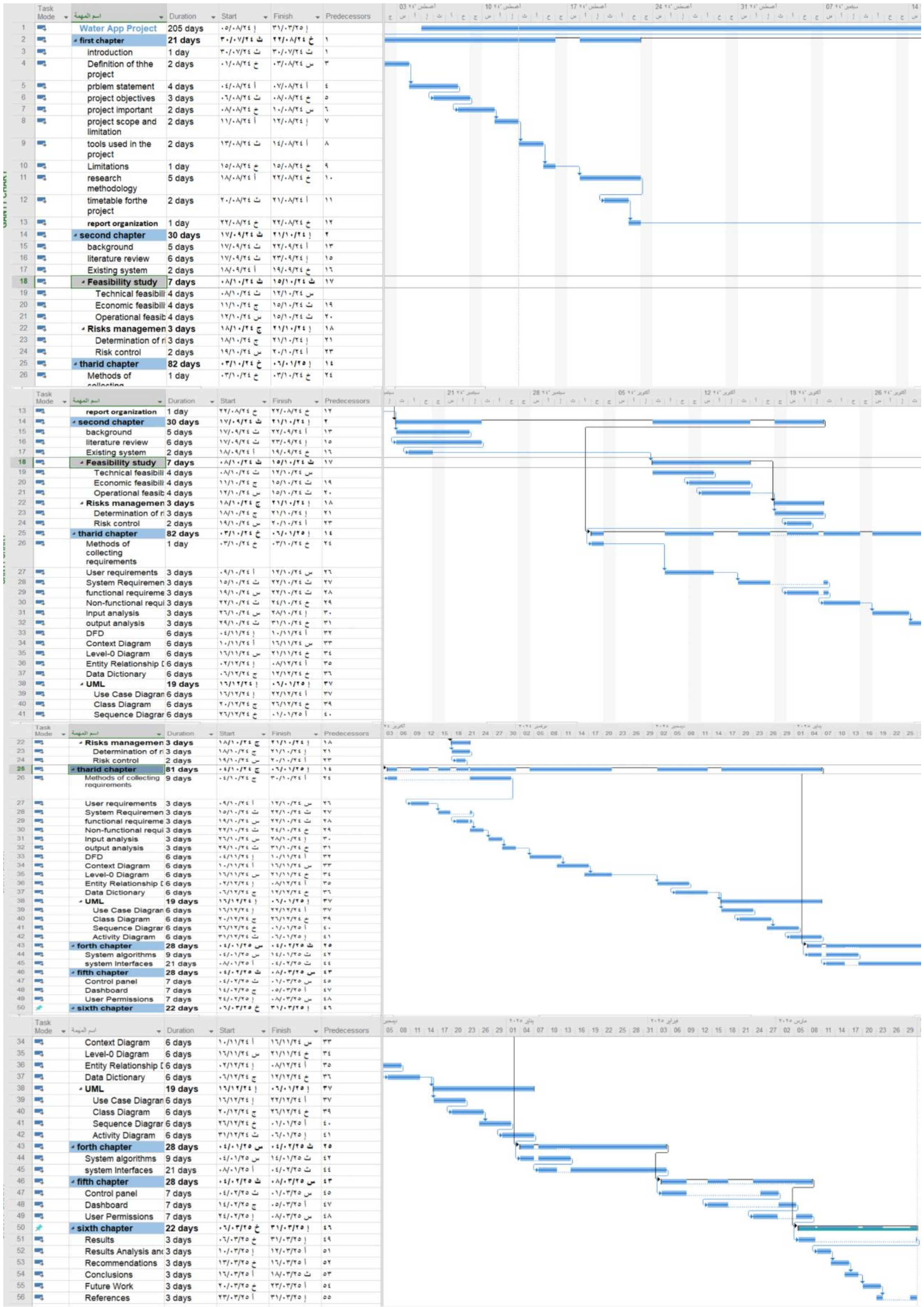
Project Timetable:

Month	Sept-2024				Oct-2024				Nov-2024				Dec-2024				Jan-2025				Feb-2025				Mar-2025			
Tasks	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project Identification																												
Preliminary study																												
Detailed study																												
Analysis																												
Design																												
Implementation																												
Evaluation and testing																												
Documentation																												

3Table 1-2(table of time)

1.10 Gantt chart for the project

Project planning is a discipline addressing how to complete a project in a certain timeframe, usually with defined stages and designated resources. One view of project planning divides the activity into these steps: setting measurable objectives, identifying deliverables, and Scheduling. We have divided our project plan to five main stages each one has different subtasks; each will be assigned to each member of the team as shown in figure (1-2).



2Figure 1-2 (Jant digram)

1.11 Report Organization

- **Chapter One: Introduction**

This chapter introduces the Water Services System project, outlining its purpose and significance. It defines the project's goals, identifies the problems it aims to address, and explains the importance of automating water service processes. Additionally, it covers the project's scope, limitations, and the tools used for development.

- **Chapter Two: Background and Literature Review**

This chapter provides context for the project by examining existing systems and relevant literature. It discusses the challenges faced in water service delivery, particularly in regions with limited access. The chapter also highlights global and local systems, comparing their features to the proposed solution and conducting a feasibility study.

- **Chapter Three: System Analysis**

This chapter details the analysis of system requirements, including user needs and functionalities. It presents various diagrams, such as Data Flow Diagrams (DFD) and Entity Relationship Diagrams (ERD), to illustrate data interactions and processes. The chapter concludes with a discussion of system requirements and the methodologies employed for requirement collection.

- **Chapter Four: Design**

This chapter outlines the design phase of the Water Services System, focusing on algorithms and system interfaces. It describes the login process, water request handling, order management, and notification systems. The chapter emphasizes the importance of user-friendly interfaces and efficient algorithms to optimize service delivery and enhance user experience.

- **Chapter Five: System Implementation**

This chapter discusses the implementation of the Water Services System, detailing the steps taken to build and deploy the application. It covers the technologies used, coding practices, and the integration of various system components. Challenges encountered during implementation and how they were resolved are also addressed.

- **Chapter Six: Results and Discussions**

This chapter presents the outcomes of the project, evaluating the effectiveness of the implemented system. It discusses user feedback, performance metrics, and the overall impact of the application on water service management. The chapter also reflects on lessons learned and potential areas for future improvement.

2. Background and Literature Review

2.1 Background

Yemen is considered one of the world’s most water- scarce countries. About 18 million people lack access to safe water and sanitation, and providing water will likely be one of the biggest problems that people will encounter in coming years. Complicating the issue is the fact that conflict has had a severe impact on water infrastructure. Average annual rainfall varies, with droughts already affecting some areas.

So, during that time people started to transport water manually in containers from wells be it far or close to the area they live in, sometimes in multiple rounds, to provide the bare minimum water needed for everyday life which was taxing in both effort and time. After using that method for quite a while people started combining water tanks and trucks for the purpose of transporting huge amounts of water in way less time with a specific fee. And with the revolutionizing of cell phones this market flourished even more due to how easier it got to contact one, though that still had problems regarding the miscommunication and misunderstanding that happened from a time to time with paper work of calculating and managing getting more tedious and redundant. And that's exactly where our app comes in to organize and automate all of these tasks for a smoother and more reliable experience [3].

2.2 Literature Reviews

2.2.1 Global systems

1. EyeOnWater

3Figure 2-1 (EyeOnWater)

EyeOnWater is a cloud-based application accessed through a standard web browser or through our smartphone applications. Internet access is required. Your user login credentials provide secure access to your water consumption information. When creating your account, you’ll need your water service Account ID (some utilities call them Customer Numbers). You’ll find it on you water bill. If your ID includes hyphens, trailing or leading zeros and non-numeric characters (letters) [4].

- **Advantages:**
  - ❖ Easy to learn and interact interface.
  - ❖ Provides leaks detection based on the difference of usage.
  - ❖ Implements simplistic graphs to show the flow of water.
  - ❖ The ability to set an alarm if the water usage surpasses a point set by the user.
  - ❖ Setting an email to send any notification to keep an eye on any changes.

- ❖ Provides a dashboard to keep track of the water usage in the past 7 days, month, or any specified time in detail.
- ❖ Shows when and will the counter will be read.
- ❖ The freedom to pick any period of time to print a bill of.

- **Disadvantages:**
  - ❖ Overly focused on personal use over business look.
  - ❖ Redundancy of some of the app functions.
  - ❖ The overlay can be too simplistic.
  - ❖ The graphs can be confusing to read.

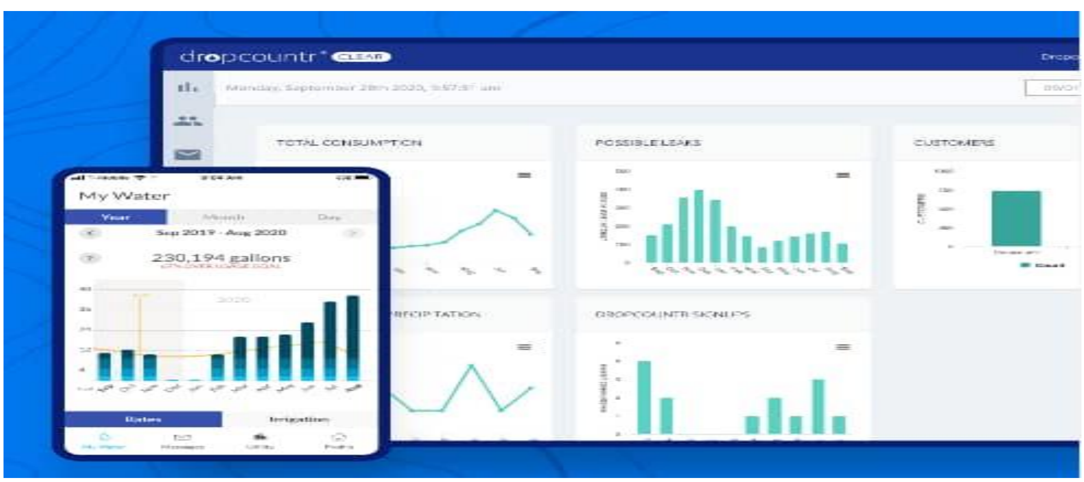
2. Tridens

4Figure 2-2 Tridens

Tridens Monetization is a proven top-performing Meter-to-Cash cloud billing software for Energy & Utility industries worldwide. Perfect for metered and nonmetered billing, complex price plans, dynamic tariff models, and customer management [5].

- **Advantages:**
    - ❖ Reduce cost-to-serve and deliver service excellence.
    - ❖ Sell more and Accelerate time-to-market.
    - ❖ Set up complex price plans and dynamic tariff models with personalized offers, promotions, discounts, contracts, and terms.
    - ❖ Maximize revenues with pre-integrated payments, receivables, and real-time analytics. Realize faster ROI and lower TCO with a standards-based, scalable cloud-native architecture powered by Amazon Web Services.
    - ❖ Reports, Dashboards, and Real-time Analytics.
    - ❖ Analyze usage patterns, revenue trends, and customer behavior.
    - ❖ Support any utility service and business model in a configurable environment.
- **Disadvantages:**
    - ❖ The subscription costs could be heavy.
    - ❖ Getting used to it could take quite the time.
    - ❖ More towards bigger businesses and decision making.
    - ❖ Interfaces are more complex

3. Dropcountr



5Figure 2- 3 Dropcountr

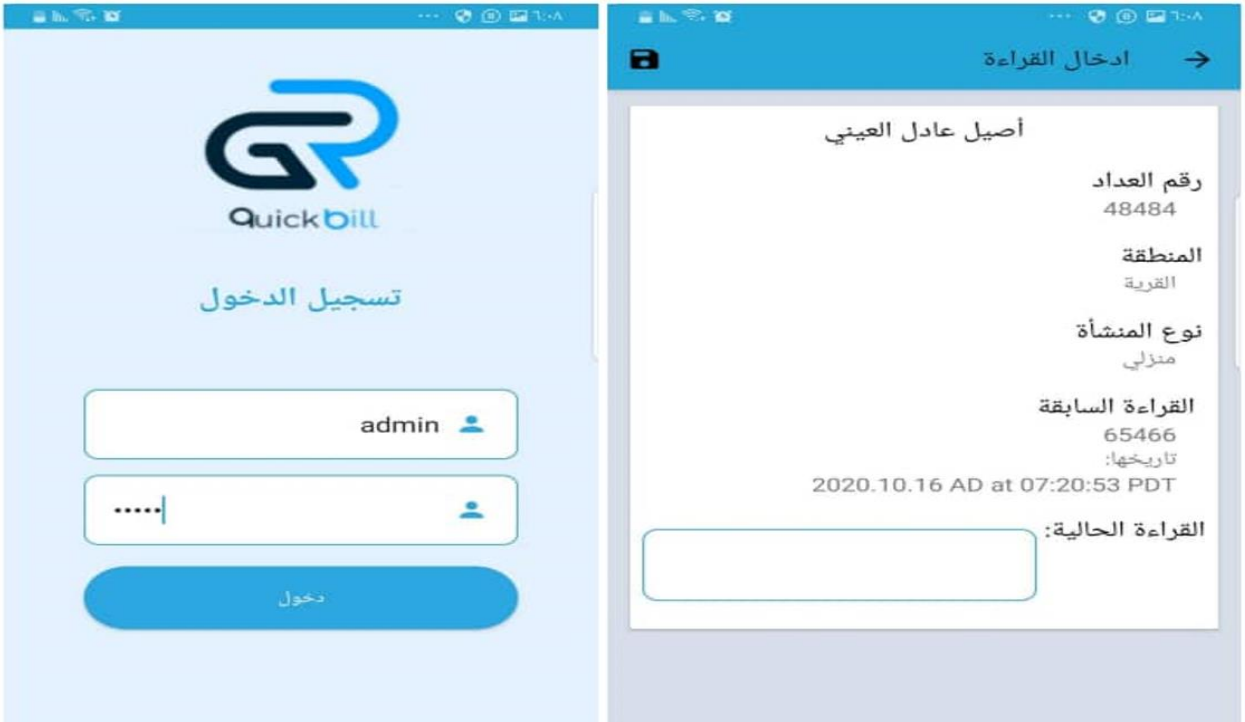
Cloud-based data analytics and customer engagement. Dropcountr translates data generated by water meters into actionable information for utility staff and their customers [6].

- Advantages:
  - ❖ The easiest one out of the three to use.
  - ❖ Automate alerts and transition to digital communications.
  - ❖ Reduce customer service call volume.
  - ❖ Multi-platform.
  - ❖ Match water use to bill and perform self-service tasks like electronic bill payment and start/stop service.
  - ❖ Empowers your customers with information and tools, which means fewer service inquiries and questions.
- Disadvantages:
  - ❖ Is not as versatile as the other two.
  - ❖ Has less tasks than its counterparts.
  - ❖ Is not as precise with the data extracted (only years' worth).

2.2.2 Local app

- ❖ Some errors that leads to graphs not showing from time to time.

1. Quick Bill System



6Figure 2- 4 Quick Bill

It is an android app developed by a group of Yemeni students to make the process of reading counters and issuing bills easier and more manageable by utilizing the environment of smart phones. The inspiration came from the tedious and redundant methods being used by companies and the government alike in Yemen which requires a lot of physical labor and paper work for a way too little progress. All of that highlighted the need of some way of automation in this field. So to put it in perspective, Quick Bill is an Android application capable of meeting major needs of small and large electricity stations , and enables electricity companies to perform multiple operations inside and outside the company easily and swiftly, including managing counters and subscribers, issuing bills and issuing detailed reports quickly and accurately In addition to taking and uploading readings through the Android application in a seamless manner, it also offers Services for subscribers, including sending bills and notices via SMS messages .

- Advantages:
  - ❖ Bills issuing process is faster and more accurate.

- ❖ Readings, bills, and notifications are sent automatically via SMS messages.
- ❖ Reports are more detailed and precise, and getting them is faster and easier.
- ❖ Provides the company and management end with the necessary tools to monitor and control the flow of work too.

- Disadvantages:
  - ❖ The interfaces are too simplistic.
  - ❖ The operations are too basic and only provides the bare minimum results.
  - ❖ Down-time could be longer than wanted which could ruin the process of work.
  - ❖ The flow of the whole thing could be hard to learn for new users.

- We come up with a summary for the previous literature review, also comparing the features of them with our proposed system as shown in the table (2-1).

No	The system/app	Platform/s	Ease to learn and use	The ability to pay through it	Compatibility with water tracks
1	**EyeOnWater**	• Android • IOS	Relatively easy to learn and master.	Not available	Not compatible
2	**Tridens**	• Android • IOS • Website	Could be intimidating to new users with all the options.	Available	Not compatible
3	**Dropcountr**	• Android • IOS • Website	Is on the easier side of the spectrum.	Not available	Not compatible
4	**Quick Bill System**	• Android	Is so easy to use and get used to.	Not available	Not compatible
5	Our system	• Android • IOS • Website	It's so easy to navigate, learn, and get used to for new and old users alike.	Available	Compatible

4Table (2 - 1) literature review summary

2.3 Existing system

Our app nullifies the payment problem by using multiple method. For instance, the user could either pay by cash when the track driver gets to their location, starting a tab to pay later when they could, or through an electronic app just to make the app the more convenient to the user. As for the water tracks, since the government water supply is not consistent or even non-existent which calls for the need for water tracks which are the focus of our app to make the access to them easier and faster, and on the topic of the situation in our country our app also solves the mineral levels in water problem by showing the user which well has the higher percentage and gives them the choice to pick which one is to their preference.

2.4 Feasibility Study

A feasibility study is a comprehensive evaluation of a proposed project that evaluates all factors critical to its success in order to assess its likelihood of success. A proposed plan or project is evaluated for its practicality. As part of a feasibility study, a project or venture is evaluated for its viability in order to determine whether it will be successful.
As the name implies, a feasibility analysis is used to determine the viability of an idea, such as ensuring a project is legally and technically feasible as well as economically justifiable. It tells us whether a project is worth the investment—in some cases, a project may not be doable. There can be many reasons for this, including requiring too many resources, which not only prevents those resources from performing other tasks but also may cost more than an organization would earn back by taking on a project that isn't profitable.

2.4.1 Technical Feasibility

A. Needed Hardware

The hardware will be needed are computer with at least CPU core i5 RAM at least 8GB.

B. Needed Software

Software	Usage
Windows 10 or 11	To run the needed environments and compliers.
Android studio	To code and compile
Visual studio code	To code and compile
Flutter SDK 3.34	The flutter platform for coding

5Table (2 - 2) Needed Software

C. Is it practical?

Yes, it is practical, since all needed hardware isn't that costing to provide.

D. Needed Expert

The user should know how to use phone at least.

2.4.2 Economic Feasibility

Requirements	cost per piece	Quantity	total cost
Windows 10	30$	5	150$
PC	750$	2	1500$
Printer	100$	1	100$
Hard Drive	70$	1	70$
Total cost:	1820$		

6Table (2 - 3) Financial Feasibility

2.4.3 Operational Feasibility

The system performance tested according to the following items in terms of the operational feasibility

i. Performance

Offers a smoother and a far faster experience through the entire water ordering and supplying processes

ii. Information

1. Input Data:

   The data inserted should be comprehensible and meaningful through the use of data validation.

2. Stored Data:

   Data must be stored in a neat and readable fashion while making sure of its accuracy.

3. Output Data:

Data retrieved must be organized and in a form that’s easy to understand even by non-tech savvy users.

iii. Economic

1. Profit:

Minimize the cost of calls and transportations to talk the assigned person in person.

Facilitate the task of monitoring who got supplied by the water, when, and how much they took, which save time and effort.

iv. Efficiency

- ❖ Waste time:

The app will provide a way of keeping eye on the whole processes happening which will reduce the time of users coming in person and asking questions.

- ❖ Effort:

There won't be any difficulty in using the app whatsoever since the interface is easy to understand and master.

- ❖ Required materials:

Mentioned above in economic feasibility.

v. Services

1. Ease to learn:

The flow of the app is easy to keep track of even for new users and doesn’t require any former knowledge.

2. flexibility:

The app is flexible and can deal with errors easily be it from front or back end.

3. compatibility:

The app is compatible with android system smart phones.

2.5 Risk Management

2.5.1 Determination of risk

A. Qualitative Assessment

- ❖ Risk Matrix: Create a matrix to categorize risks by their likelihood (low, medium, high) and impact (low, medium, high).
- ❖ Prioritization: Focus on high-likelihood and high-impact risks first for mitigation efforts.

B. Quantitative Assessment

- ❖ Data Analysis: Use historical data to estimate the probabilities and impacts of specific risks.
- ❖ Monte Carlo Simulation: A statistical technique to model the probability of different outcomes based on variable inputs.

2.5.2 Risk Control

A. Mitigation Strategies

1. **Technical Solutions**

- ❖ Redundancy: Design systems with backup components to ensure continuity.
- ❖ Regular Maintenance: Implement a preventive maintenance schedule to reduce the risk of equipment failure.

2. **Environmental Management**

- ❖ Sustainability Practices: Use water conservation techniques to reduce vulnerability to climate variations.
- ❖ Emergency Response Plans: Develop and regularly update response strategies for natural disasters.

3. **Regulatory Compliance**

- ❖ Legal Consultation: Engage legal experts to ensure compliance with current laws and anticipate changes.
- ❖ Stakeholder Engagement: Involve stakeholders early in the planning process to address concerns proactively.

B. Transfer Strategies

- ❖ Insurance: Obtain insurance for natural disasters, equipment failures, and liability coverage.
- ❖ Contracts: Use performance bonds and warranties in contracts with vendors and contractors.

C. Acceptance Strategies

- ❖ Risk Tolerance Assessment: Define acceptable levels of risk for the project and communicate this to stakeholders.
- ❖ Contingency Plans: Establish contingency funds or plans for risks deemed acceptable but still possible.

3. System Analysis

3.1 Methods of collecting requirements

- **Interviews**

As a part of our data collecting process, we have visited two of the most renowned wells in Sana’a, AL-Abhar Well and Bi'r Ash Shaif. In both sites we were faced with a warm welcome followed by some really enthusiastic replies from the well owner. After explaining the idea of our app, they seemed invested in the whole process and talked to us about the problems they are facing through their job such as keeping track of the whole billing and archiving consumers data, and being clear to the consumers about their readings in real time. Another point that seemed to grab their interest was the idea of paying through the app since they had a lot of problems collecting the cash directly from the consumers which they said would be a real-life saver. Lastly, we discussed with them using the barcode tech to make the whole checking and writing down the reading process easier and more efficient, which also has been welcomed by them.

It was definitely a beneficial and reassuring trip that gave us a different point of view and granted us a whole new perspective on our future view on the app, and reassured us in how accepted would our app be.

  - Some of the questions that is done in the interviews are:
    - Could you please explain the order and steps of your process?
    - How do you deal with archiving the consumers usage data?
    - Do you use the data gathered from the consumers usage to improve your work?
    - How do you organize the flow and keep the water providers in check?
    - What are the difficulties that face you on a regular basis and in need for automation in your opinion?
    - Is there any suggestions or recommendations you would like to add to our app and similar ones?

- **Observation**

To use this method, we went to multiple well owners and water providers. We observed how processes and activities are done. According to this observation, we have obtained a lot of ideas and knowledge that were not clear from the beginning.

Some of these problems that occur with the water provider are: first, the inability of the consumer to quickly access the water provider; second, the inability of the consumer to choose the right size and price according to his needs; third, the delay of the water supply for the consumer due to work pressure and lack of arranging tasks.

Some of these problems are related to the owner of a well: first, the water was cut off without warning; second, late payment due to not receiving the invoice at the right time. There are things that happen between the owner of the well and the water supplier: first, the frequent crowding of water suppliers; second, not knowing the exact number of times the provider provides.

- We will divide the problems facing all parties:

a) Well owner and consumer:

The consumer registers and pays the service fee to bring water from the well to the house, then the owner of the well sets the unit price for the customer on a monthly basis, after that, the owner of the well records the readings at the end of the month and the water meter issues monthly bills to the consumer.

- Problems while providing water supply service include:

1. The delay in getting the service due to lack of communication.
2. Delay in delivering invoices on time.
3. Late payment.
4. Some errors occur in the reading of the water meter and the method of calculating and issuing the bill.

b) Well owner and water provider:

The water provider fills the water from the well; then the owner of the well records the number of times the provider gets water from the well and the cost in two ways, either immediately or at the end of the day, where prices vary according to the sizes and capacities of the water providers and the well.

- The problems that occur include:

1. Not adjusting the prices of filling from the well to the owner of the water provider.
2. It is not easy to record the number of filling times for the water provider on a daily basis.

3. There is no regulation of payment methods, whether cash or deferred.

c) Water supplier and consumer:

The consumer submits the request by calling the water provider, where the water provider comes to empty the water at the consumer's house; then the consumer pays the cost to the water provider through three methods: (direct payment (cash), payment at the end of the month (duration), payment by money order)

- The problems that occur include:

1. It is not easy to reach the water provider by calling.
2. It is not easy to explain the site to the water provider.
3. There is no organization and arrangement in the tasks of the owner of the water provider.
4. There are problems in the payment methods by the consumer.

3.2 User requirements

1. The ability to subscribe to the application via the phone number.
2. Order water provider or subscribe to the well through the app.
3. Determining who paid and who did not without resorting to paperwork.
4. Issuing invoices periodically.
5. The ability to read the water meter via a barcode.
6. Determine the location of the consumer, the Wells and through the map.
7. Knowing the working hours of the well owners and the Water supplier and whether it is active or not.
8. Receive notifications through the app.

3.3 System Requirements

1. Multi-Role Authentication System
2. Barcode Scanning Functionality
3. Real-Time Order Management

4. Automated Reporting Engine
5. Location-Based Services
6. Offline-Capable Architecture
7. Multi-Platform Support
8. Notification System

3.4 Functional Requirements

1.1 The ability of each user to register with the application, whether as a consumer or a water distributor.

1.2 Enabling the user to register using the phone number and receive confirm SMS message to enter the app.

2.1 Enabling the consumer to send order to the water provider and selecting the unit that he wants.

2.2 Also enabling the user to send a subscription request in the well through the app.

3.1 Determine the consumer who paid cash.

3.2 Determining who took a deferred account and the account at a later time, specifying the period.

4.1 When reading the consumer meter, the system issues the current readings with the previous readings, and determines the difference through an invoice.

4.2 The invoice includes the total cost to the consumer, specifying if there are any arrears.

4.3 Issuing invoices periodically or on a date specified by the user.

5.1 Instead of paper transactions, the application will read the meter by scanning the barcode on the meter.

5.2 When scanning the barcode, the consumer ID is entered and the required data, such as cost, are shown.

6.1 The application shows the consumer's location to enable access to the water resource without contacting the consumer.

6.2 The application also informs the consumer of the location of the water supplier closest to him and the distance he needs to reach him.

6.3 He also knows the location of the owner of the well.

7.1 The application enables the consumer to know the working hours of wells and water suppliers by sending notification.

7.2 Order water in a timely manner.

7.3 It also enables the consumer of the active wells and water providers at the time you open the application.

8.1 The application will confirm the consumer of any filling process through notifications.

8.2 The well's owner will be able to send any new information or warning through the notifications.

3.5 Non-Functional Requirements

Non-functional requirement are requirements that are not directly concerned with the specific services delivered by the system. They may relate to emergent system properties such as usability, dependability (Reliability, Validation, Safety, Security, and Performance). Alternatively, they may define constraints on the system implementation such as the capabilities of Input/output devices or data representation used in interface with other applications.

II. Usability

- The user interface must be easy and attractive to use.
- The system is expected to be efficient to use, which means it may take less time to accomplish the required functions.
- The overall system expected to be easy to use.

III. Dependability

a) Reliability

- The application should operate as the user requires.
- Data must be stored and retrieved in a consent way.
- The rate failure in application should be small.
- The recovery from failure should be very fast.

b) Validation

If the user has entered wrong inputs, the application informs the user about that by notification massages.

IV. Safety

The system should not be harmful for the environment and people that surrounding.

V. Security

The application has authentication method as username and password, also assess the application according to the user privilege.

VI. Performance

The application should be high performance (speed in performing tasks) to meet all the requirements. Performing the application operations and its performance is set as the following points:

a) Availability

An application's availability is the amount of time that users can use the application operations. In addition, Application can be easily accessed at any time.

b) Throughput

The application should be as fast as possible in performing processes of the application.

VII. Accuracy

The application designed to perform operations with correct output. Also, when issuing or printing reports the data entered should be correct.

VIII. Maintainability

There will be a continuously maintenance for the system, also to keep up with the new technology and changing of interfaces by time go on. In addition, the application maintainable with addition requirements and with user new or change requirements.

3.6 Input analysis

Input analysis involves identifying the data that will be required by the system to function effectively. The following inputs will be necessary for the Water Services System:

1. **User Registration Data**:
    - Phone number
    - Personal information (name, location)
    - User type selection (consumer, water provider, well owner)
2. **Water Consumption Data**:
    - Water meter readings (input via barcode scanning)
    - Units of water consumed
    - Consumer ID for tracking usage
3. **Order and Subscription Requests**:
    - Selected water provider or well
    - Number of units requested
    - Preferred delivery date
4. **Payment Information**:
    - Payment method selected (cash, deferred, electronic)
    - Transaction details for invoicing
5. **Notifications and Alerts**:
    - User preferences for receiving notifications regarding water usage and service availability

3.7 Output analysis

Output analysis focuses on the data that the system will produce and how it will be presented to the users. The following outputs will be generated by the Water Services System:

1. **User Confirmation Messages**:
    - Registration confirmation via SMS
    - Subscription request status (accepted or rejected)
2. **Water Consumption Reports**:
    - Detailed reports on units consumed over specified periods
    - Monthly or custom invoices including total costs, payment status, and any arrears
3. **Order Status Updates**:

- Notifications regarding the status of water orders (accepted, in progress, delivered)
- Alerts for upcoming deliveries or service changes

4. **System Notifications**:
   - Alerts for any changes in water availability
   - Notifications for payment reminders and confirmations
5. **Administrative Reports**:
   - Reports for administrators summarizing user activities, payment transactions, and service utilization

3.8 DFD Diagram

Data Flow Diagrams (DFD) represent the flow of information between external entities and the system. They are essential for understanding how data moves within the system and for identifying the various processes that interact with the system.

A DFD provides a visual representation of the system's processes and data flows. It helps to clarify how inputs are transformed into outputs, which is crucial for both system design and analysis. The diagram typically includes symbols for processes, data stores, external entities, and data flows, allowing stakeholders to easily grasp the system's functionality.

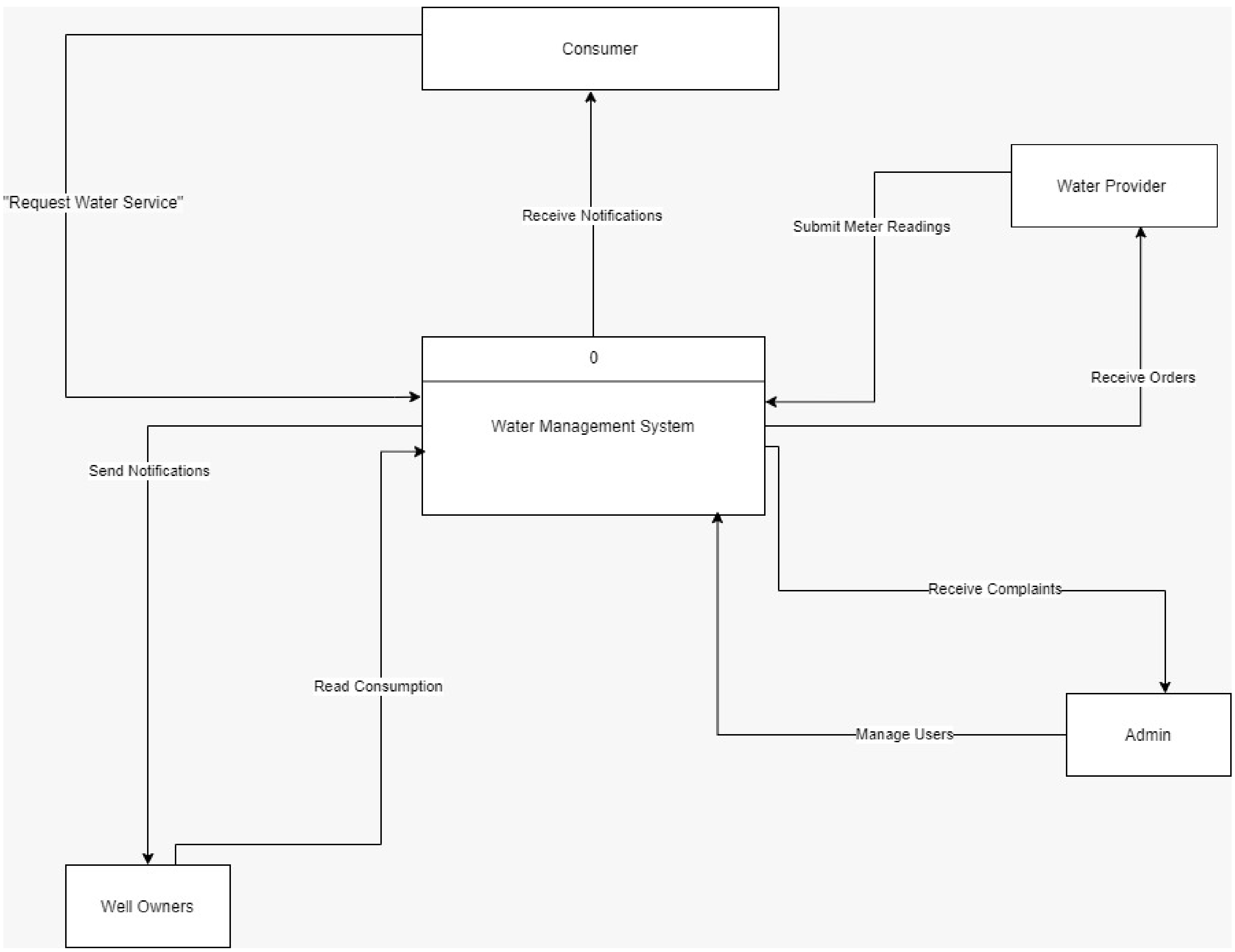
3.9 Context Diagram

The Context Diagram illustrates how external entities interact with the Water Services Management System, highlighting the flows of information between them. This diagram aims to provide a clear understanding of the system's scope and its role within the surrounding environment.

1. **External Entities**:
   - **Consumer**: Interacts with the system by requesting water services and receiving notifications related to water management.
   - **Water Provider**: Sends meter readings and receives orders and complaints from consumers.
   - **Well Owners**: Responsible for reading water consumption and sending relevant notifications to the system.
   - **Admin**: Manages users and oversees the overall operations of the system.
2. **Data Flows**:
   - **From Consumer to System**:

- Requesting water services.
- Receiving notifications.

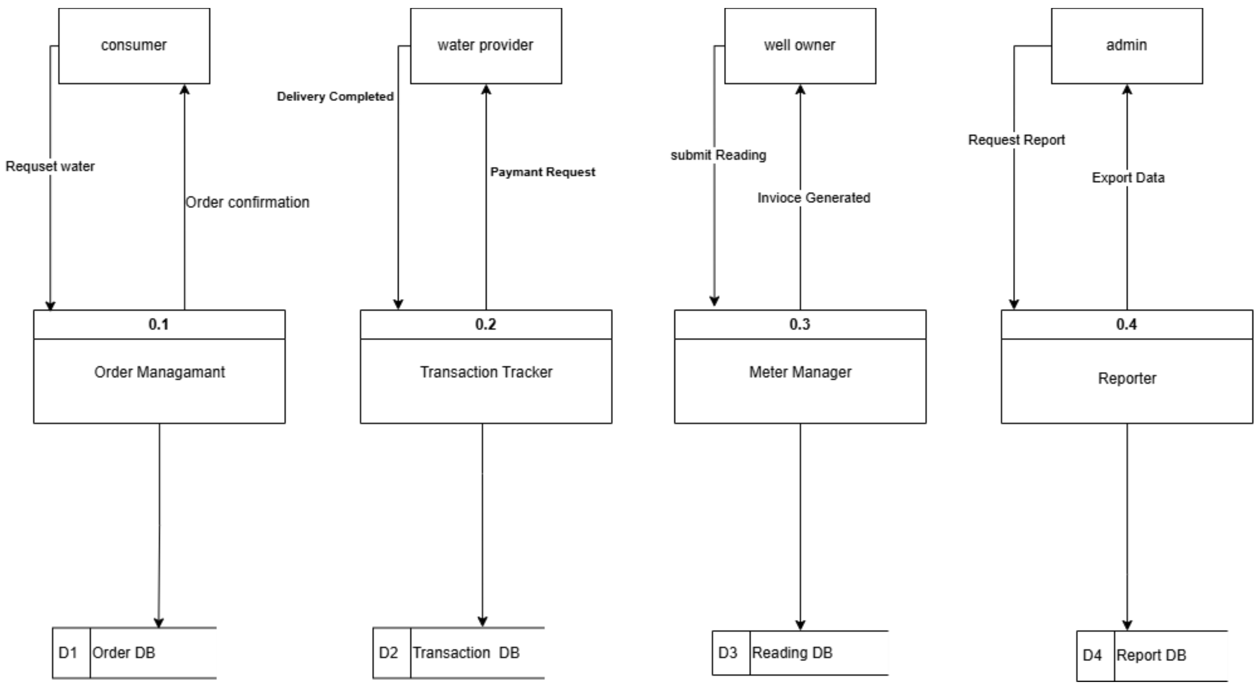
- **From Water Provider to System**:
  - Sending meter readings.
  - Receiving orders and complaints.
- **From Well Owners to System**:
  - Reading water consumption.
  - Sending notifications.
- **From Admin to System**:
  - Managing users and monitoring the system's performance.



7Figure 3-1 (Context Diagram)

3.10 Level-0 Diagram

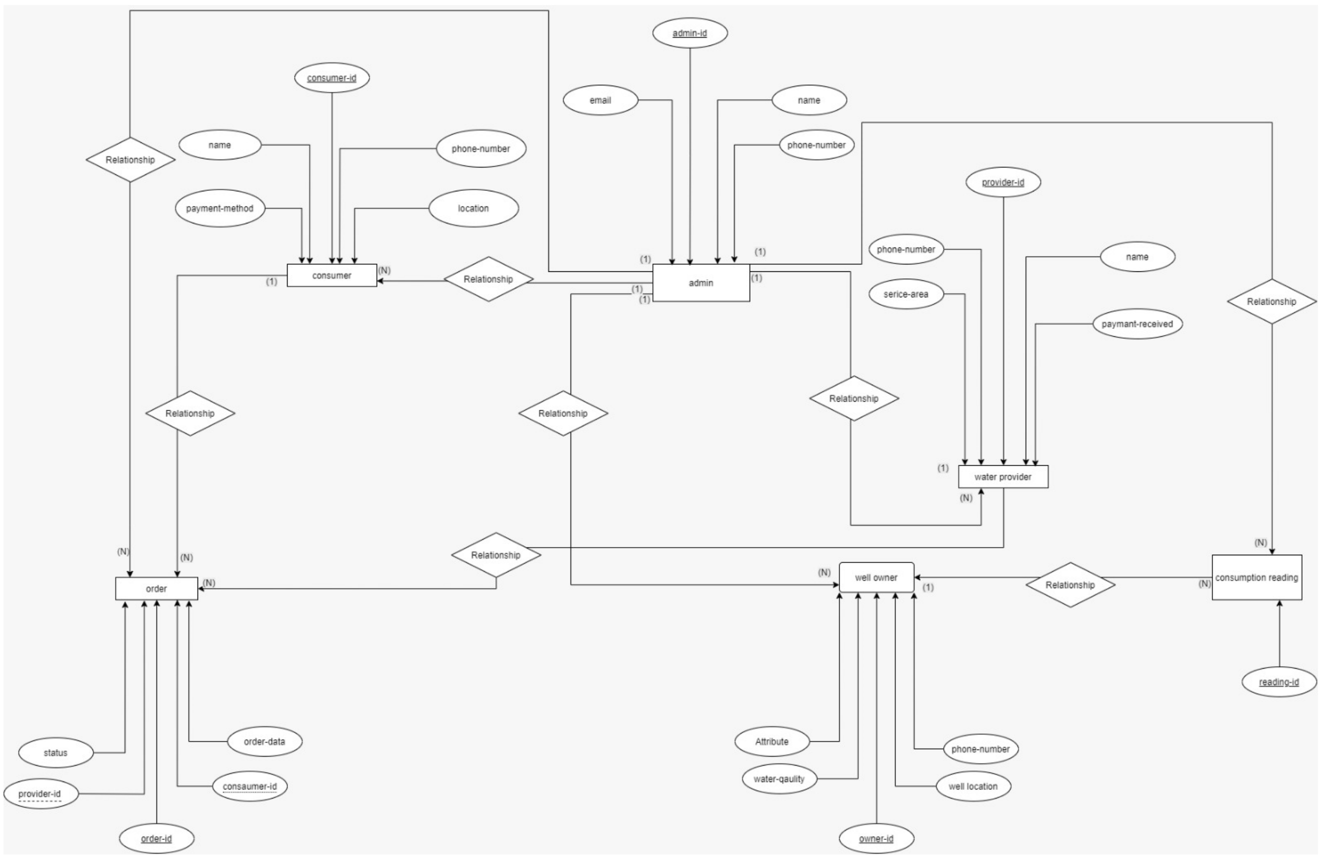
"This zero-level diagram provides an overview of the Water Services System, illustrating its core interactions with four key stakeholders: Consumers, Water Providers, Well Owners, and Admins. The system enables Consumers to request water and make payments, facilitates Water Providers in managing availability and orders, empowers Well Owners to record readings and generate invoices, and provides Admins with management and reporting functionalities."



8Figure 3-2 (Level-0 Diagram)

3.11 Entity Relationship Diagram(ERD)

Database design: ER diagrams are used to model and design relational databases, in terms of logic and business rules (in a logical data model) and in terms of the specific technology to be implemented (in a physical data model.)



9Figure 3-3 (Entity Relationship Diagram ERD)

3.12 Data Dictionary

Consumer Table

Description:

The Consumer Table contains detailed information about individual consumers of water services. It includes a unique identifier for each consumer, their name, contact number, location, and preferred payment method. This table is essential for managing consumer accounts and service delivery.

Name	Type	Attribute	PK/FK	Null
consumer_id	Integer	Unique identifier for the consumer	PK	No
name	String	Name of the consumer		No
phone_number	String	Contact number of the consumer		No
location	String	Address or GPS location of the consumer		No
payment_method	String	Method of payment (e.g., cash, card)		Yes

7Table (3 - 1) Data dic (consumer)

Water Provider Table

Description:

The Water Provider Table holds information about companies or individuals providing water services. Each provider has a unique ID, name, contact details, and operates in specific service areas. It also tracks whether payments have been received from consumers.

Name	Type	Attribute	PK/FK	Null
provider_id	Integer	Unique identifier for the provider	PK	No
name	String	Name of the water provider		No
phone_number	String	Contact number of the provider		No
service_area	String	Geographic area served by provider		No
payment_received	Boolean	Status of payment received from consumer		Yes

8Table (3 - 2) Data dic (water proivder)

Well Owner Table

Description:

The Well Owner Table contains details about individuals or entities that own wells. It includes a unique identifier, their name and contact information, the physical location of the well, and an optional description of water quality. This table supports the management of well-related services.

Name	Type	Attribute	PK/FK	Null
owner_id	Integer	Unique identifier for the well owner	PK	No
name	String	Name of the well owner		No
phone_number	String	Contact number of the well owner		No
well_location	String	Location of the well		No
water_quality	String	Description of water quality		Yes

9Table (3 - 3) Data dic (well owner)

Admin Table

Description:

The Admin Table lists the administrators managing the system. Each admin has a unique ID, name, and contact information, including email and phone number. This table is vital for managing system access and monitoring administrative activities.

Name	Type	Attribute	PK/FK	Null
admin_id	Integer	Unique identifier for the admin	PK	No
name	String	Name of the admin		No
email	String	Email address of the admin		No
phone_number	String	Contact number of the admin		Yes

10Table (3 - 4) Data dic (Admin)

Order Table

Description:

The Order Table tracks consumer orders for water delivery services. It includes a unique order ID, references to the consumer and provider, the date of the order, and its current status. This table is essential for managing and fulfilling consumer requests efficiently.

Name	Type	Attribute	PK/FK	Null
order_id	Integer	Unique identifier for the order	PK	No
consumer_id	Integer	ID referencing the consumer	FK	No
provider_id	Integer	ID referencing the provider	FK	No
order_date	Date	Date of order		No
status	String	Current status of the order		No

11Table (3 - 5) Data dic (Order)

Consumption Reading Table

Name	Type	Attribute	PK/FK	Null
reading_id	Integer	Unique identifier for the reading	PK	No

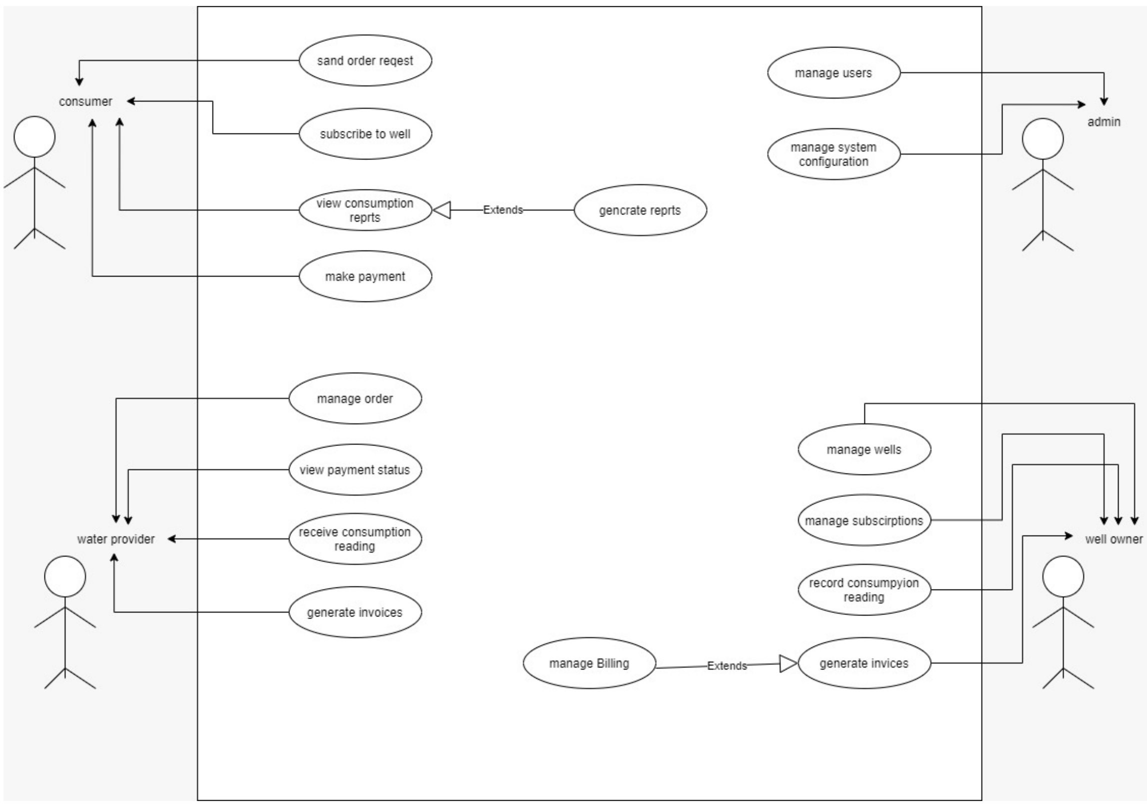
12Table (3 - 6) Data dic (consumption)

3.13 UML Design

3.13.1 Use case model

Use case model a model of how users of the system interact with the system. As such, it describes the goals of the users, the interactions between the users and the system, and the required behavior of the system in satisfying these goals.

❖ Water Service App



10Figure 3-4 (UML Design)

Use case details

Use case details explain the processes that was introduced by the use case diagram

1) The registering process

Name of the use case	Water Service Use Case
Name of the process	Registration
Actors	Well's owner, Water provider, Consumer

13Table (3 - 7) registering process

The process	• The user opens the application. • The user enters his phone number. • The application sends a verification message to confirm the phone number is belong to the user. • The user inserts his/her personal information and choose whether to be a consumer, water provider, well, or all of them, then click add.
Name of the process	Adding wells
Actors	Well owner
The process	• After registering in the app, the well owner will select subscription request. • Then he will add the wells information, such as well name, well location, units' price for water provider, units' price for consumer, and consumption price. • then the well will be added
Name of the process	Adding water provider
Actors	water provider

The process	• After registering in the app, the water provider will select subscription request. • Then he will add the vehicle information, such as vehicle name, vehicle size, units' price for consumer, and will click save. • then the water provider will be added

2) Consumer processes

Name of the process	Send superscription request
Actors	Consumer
The process	• The consumer will open the app. • Then will select send subscription request to the well. • When the well owner confirms the request • The consumer will enter the interface that show the well that he/she is subscribed to. • And will view the financial and consumption reports of each one.
Name of the process	Send water provider order
Actors	Consumer
The process	• The consumer will select request a water provider. • If the consumer wants to select a specific well, he/she is going to choose the well then, the water provider. • Else he/she will choose the water provider without selecting the well. • Then the consumer will send a request including inside it the number of units and date.

14Table (3 - 8)Consumer processes

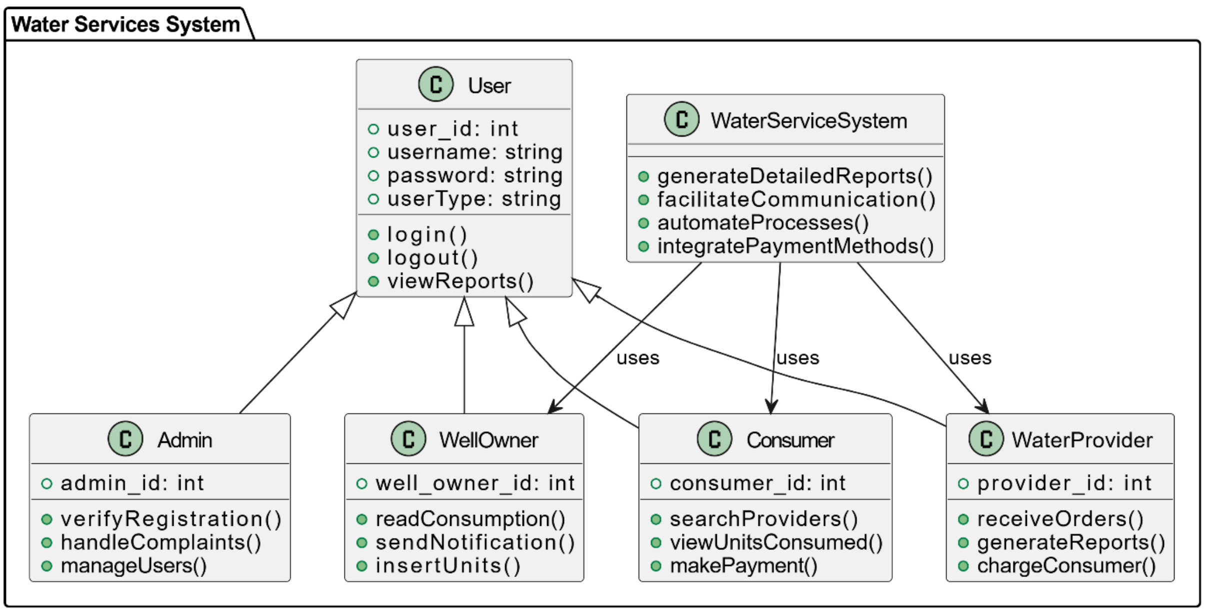
3) Well owner processes

Name of the process	Read consumption of water provider
Actors	Well owner
The process	• When the well owner wants to read the water provider readings. • He opens the interface of reading water provider consumptions. • The well owner search about the water provider by the number. • Or he scans the barcode to git the name of the water provider and vehicle. • Then the well select the units and the paid money. • Then send invoice to water provider.
Name of the process	Read consumption of consumer
Actors	Well owner
The process	• When the well owner wants to read the consumer readings. • He opens the interface of reading consumer consumptions. • Then the well owner search about the consumer by the meter number or by scanning the barcode. • Then he enters the current reading. • After that send invoice to the consumer.

15Table (3 - 9) Well owner processes

3.13.2 Class Diagram

The Water Services System is designed to automate interactions among consumers, water providers, and well owners. It includes user roles such as consumers, water providers, well owners, and admins, each with specific functionalities. Consumers can search for providers, view their water consumption, and make payments. Water providers and well owners can manage orders, generate reports, and communicate with consumers efficiently. The system facilitates seamless communication and improves overall efficiency in water service management.



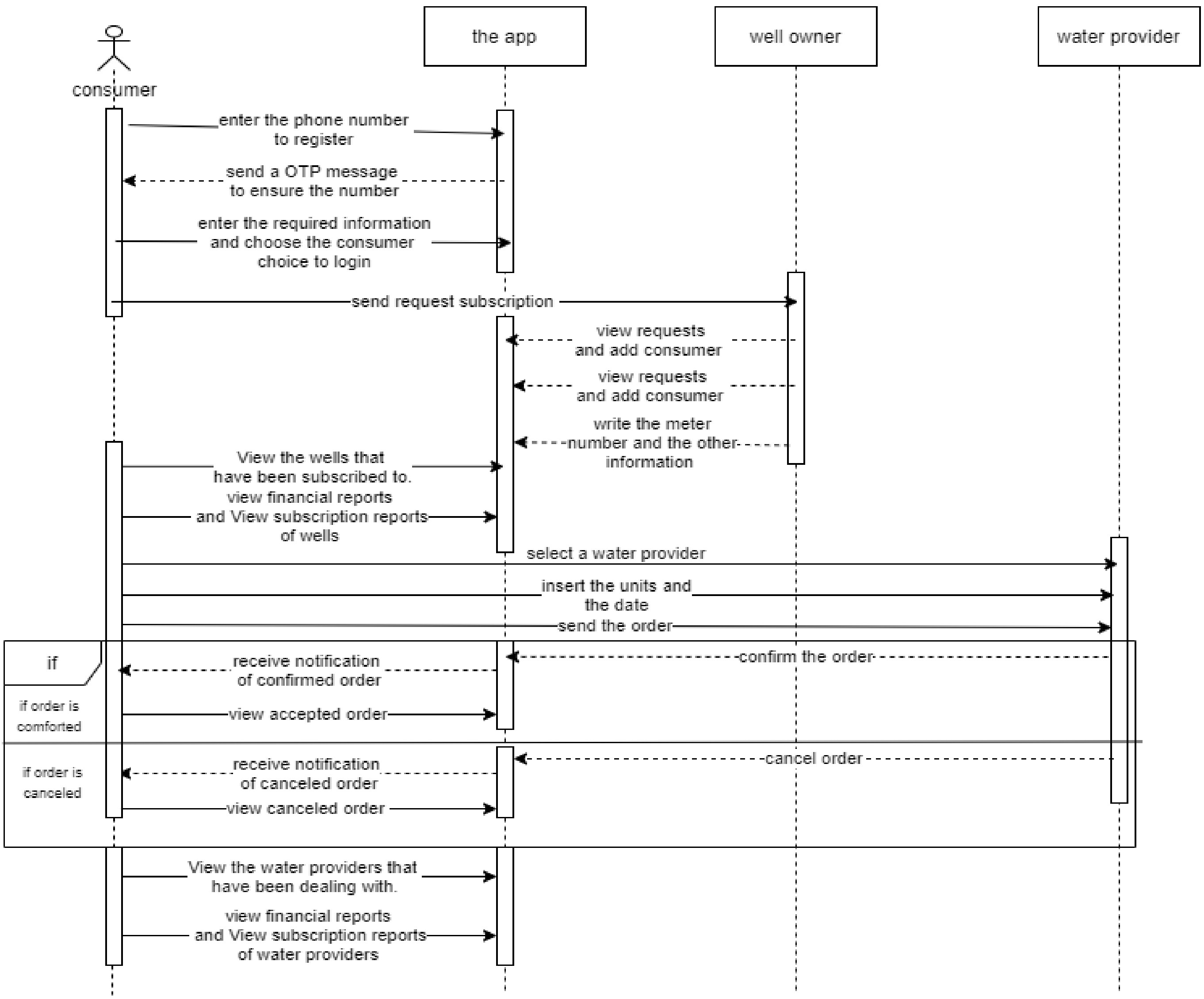
11Figure 3-5 Class Diagram

3.13.3 Sequence Diagram

A sequence diagram shows process interactions arranged in time sequence. It depicts the processes involved and the sequence of messages exchanged between the processes needed to carry out the functionality.

1) Consumer sequence diagram:

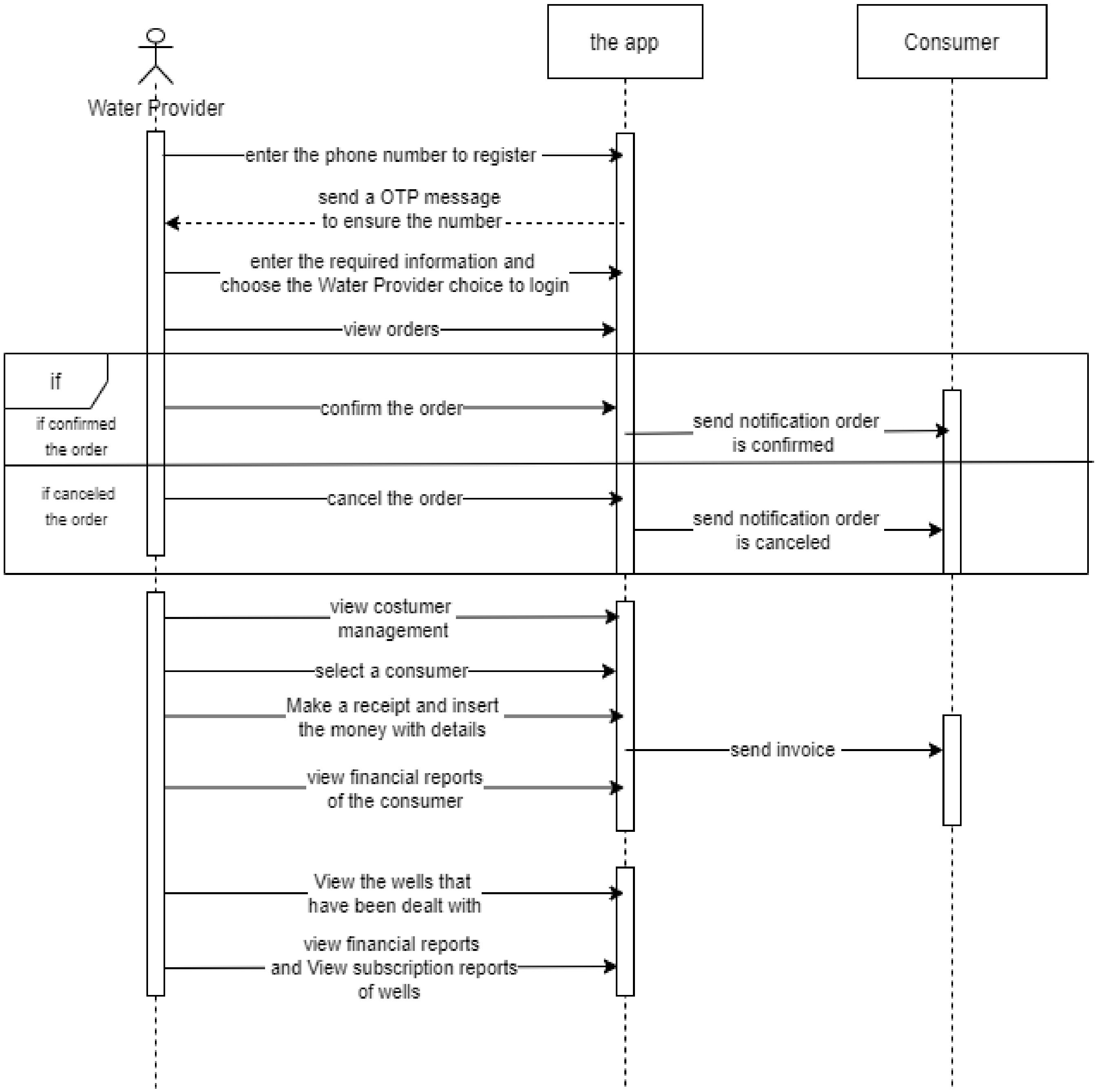
   This diagram describes all the processes that happen between the consumer, app, and other users.



12Figure (3- 6) sequence diagram (consumer)

2) Water provider sequence diagram:

This diagram describes all the processes that happen between the Water provider, app, and other users.

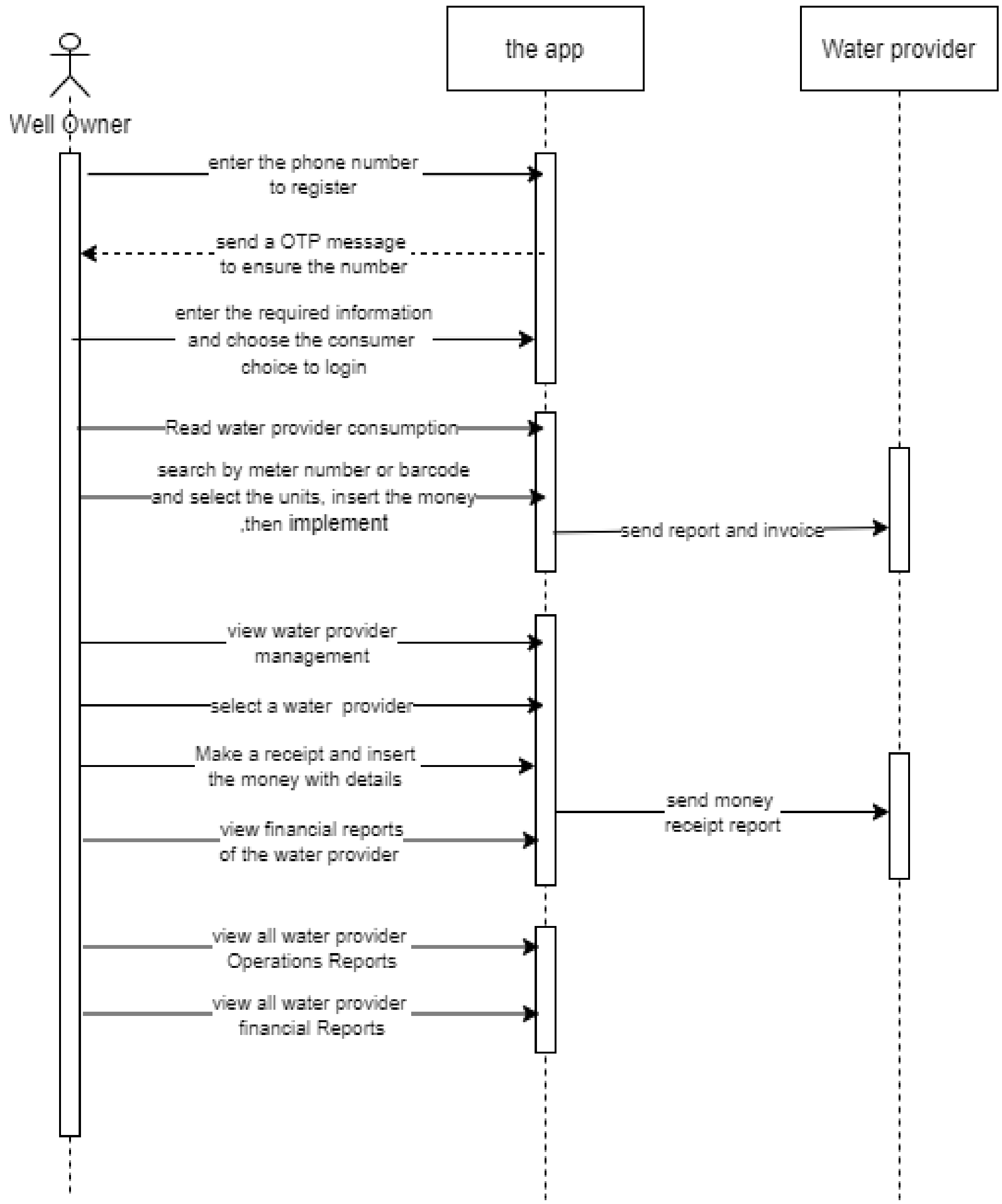


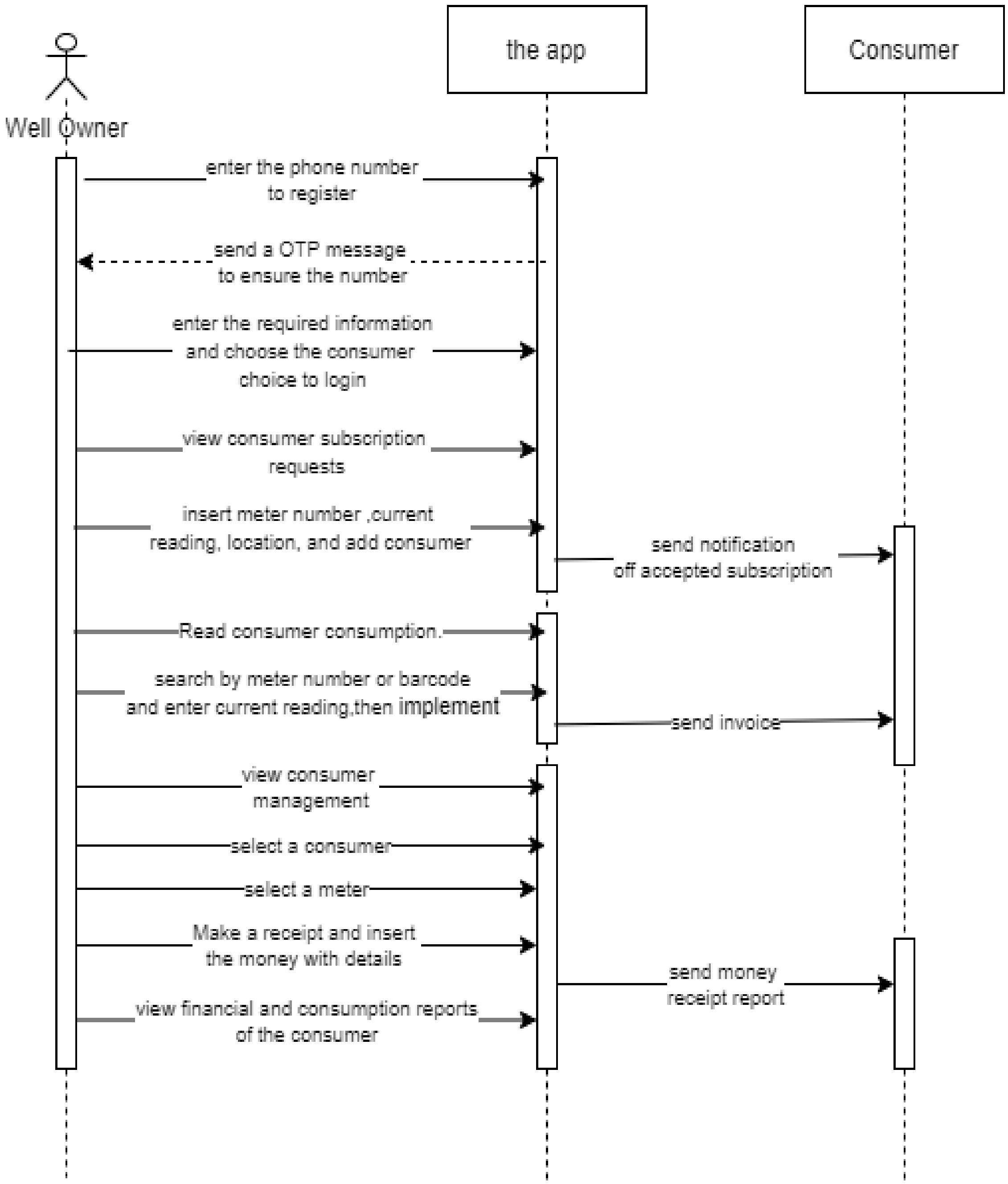
13Figure (3- 7) sequence diagram(water provider)

3) Wells sequence diagram:

This diagram describes all the processes that happen between the well, app, and other users.

Table 3-8 (wells)

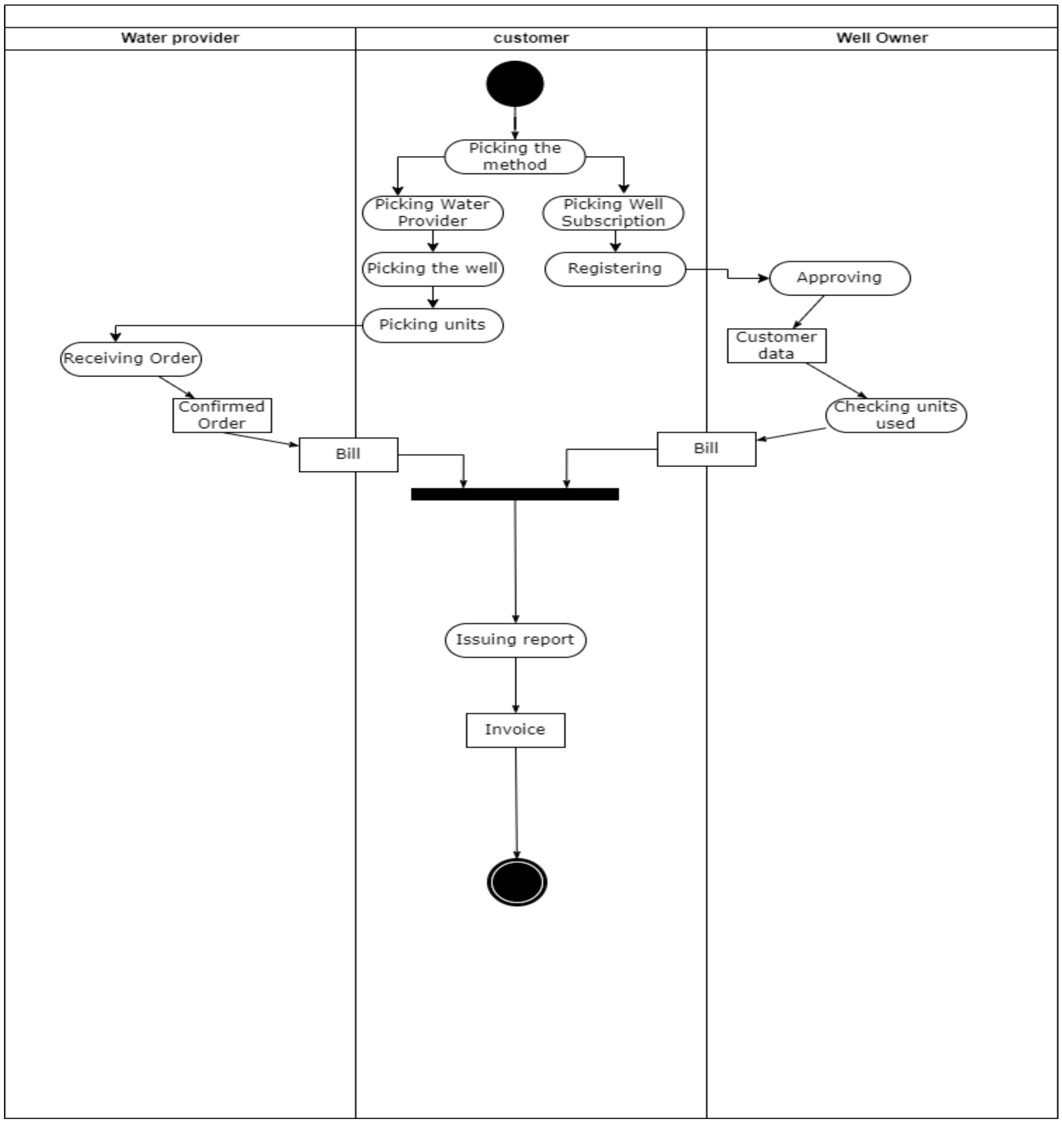




14Figure 3- 9 sequence diagram (well owner)

3.13.4 Activity Diagram

An activity diagram visually presents a series of actions or flow of control in a system similar to a flowchart or a data flow diagram.



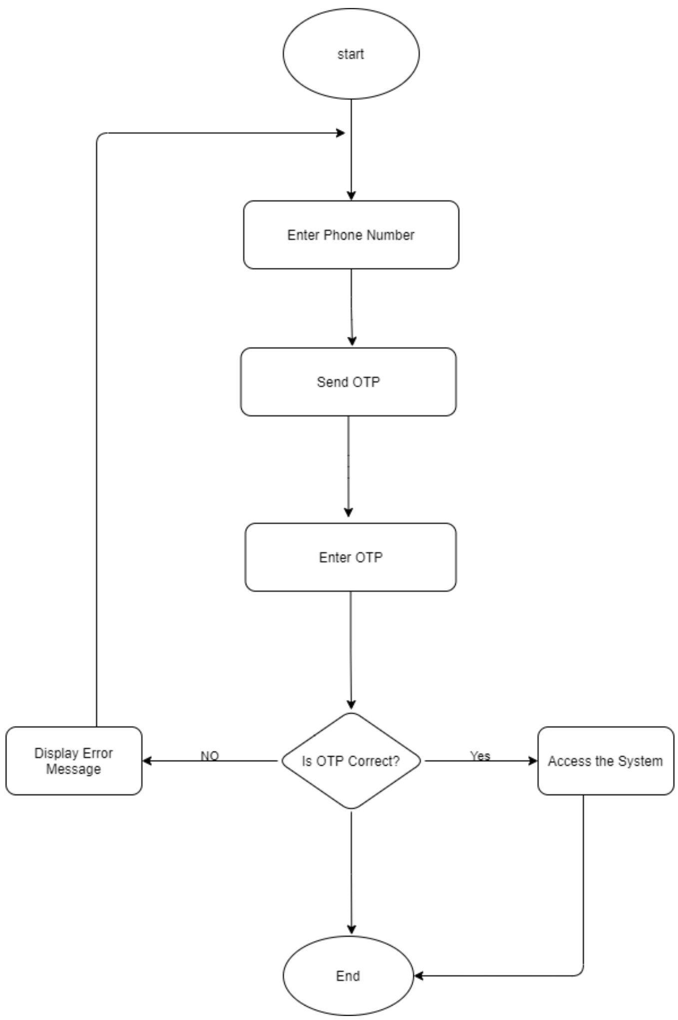
15Table 3-9 (activity diagram)

4. Design

4.1 System algorithms

4.1.1 Login algorithms

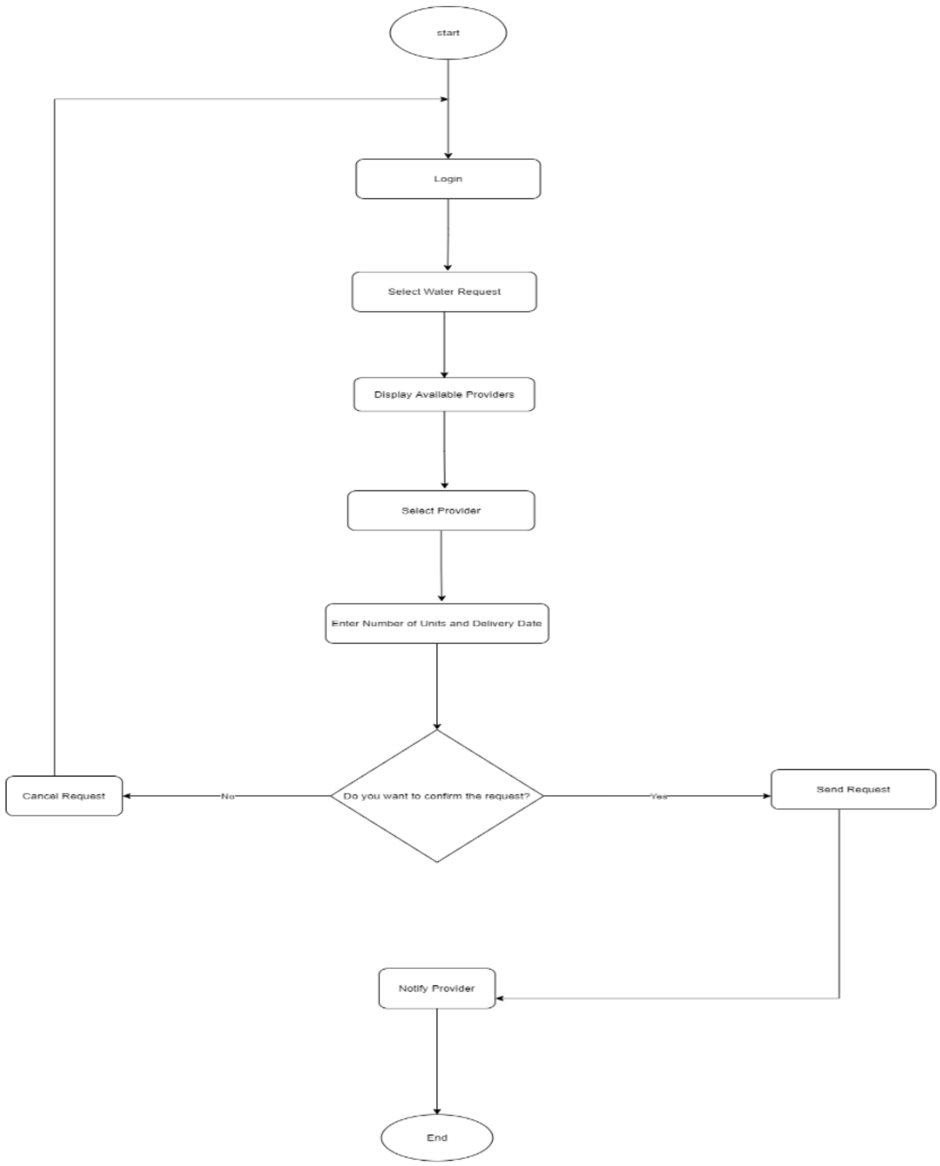
The algorithm outlines a simple authentication process for a water services application. It begins with the user entering their phone number, after which an OTP (One Time Password) is sent to that number. The user then inputs the received OTP; if it matches, access to the system is granted. If the OTP is incorrect, an error message is displayed, and the process concludes.



16 Figure 4-1 login-algorithm

4.1.2 Water request algorithm

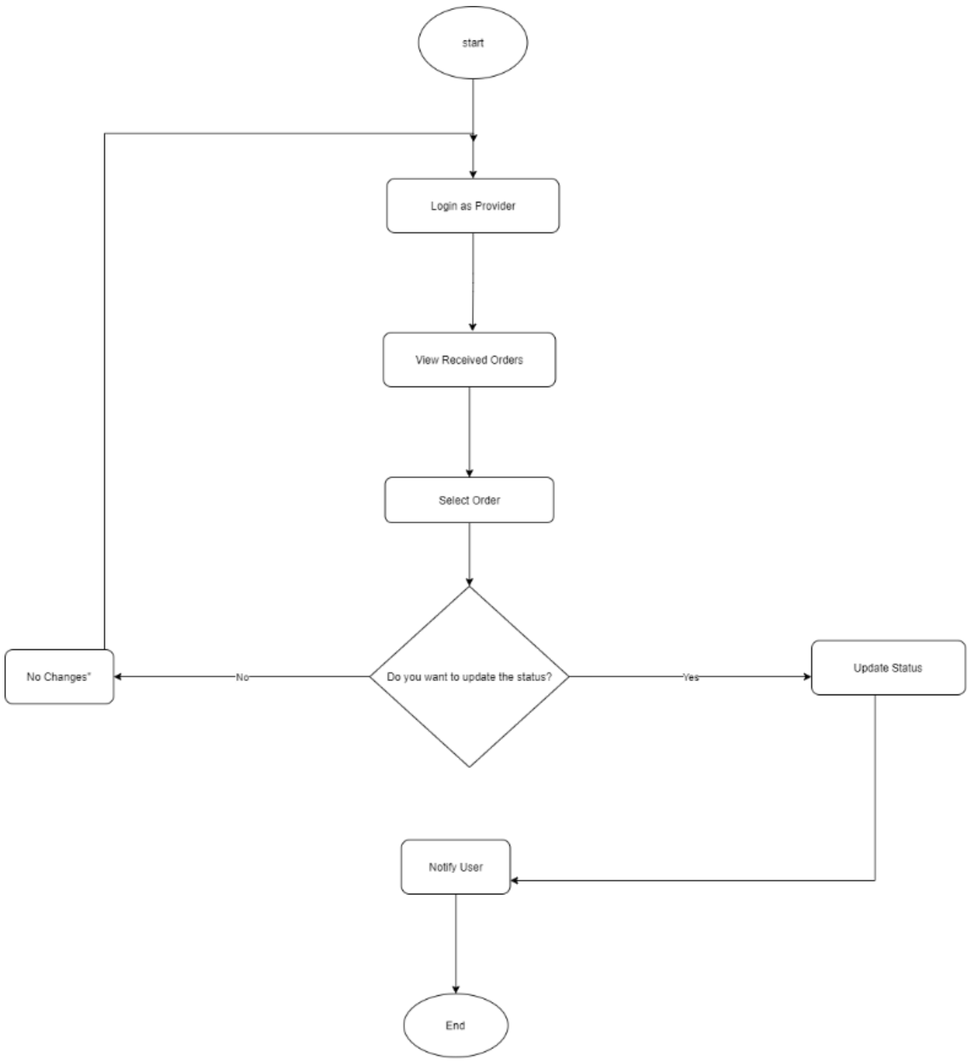
This algorithm outlines the process for a consumer to request water delivery through the application. It starts with the user logging in and selecting the type of water request. The system then displays available water providers. The user chooses a provider and enters the number of units needed along with the delivery date. After confirming the request, the system sends it to the selected provider and notifies them. The process concludes once the request is sent.



17Figure 4-2 water request algorithm

4.1.3 Order management algorithm

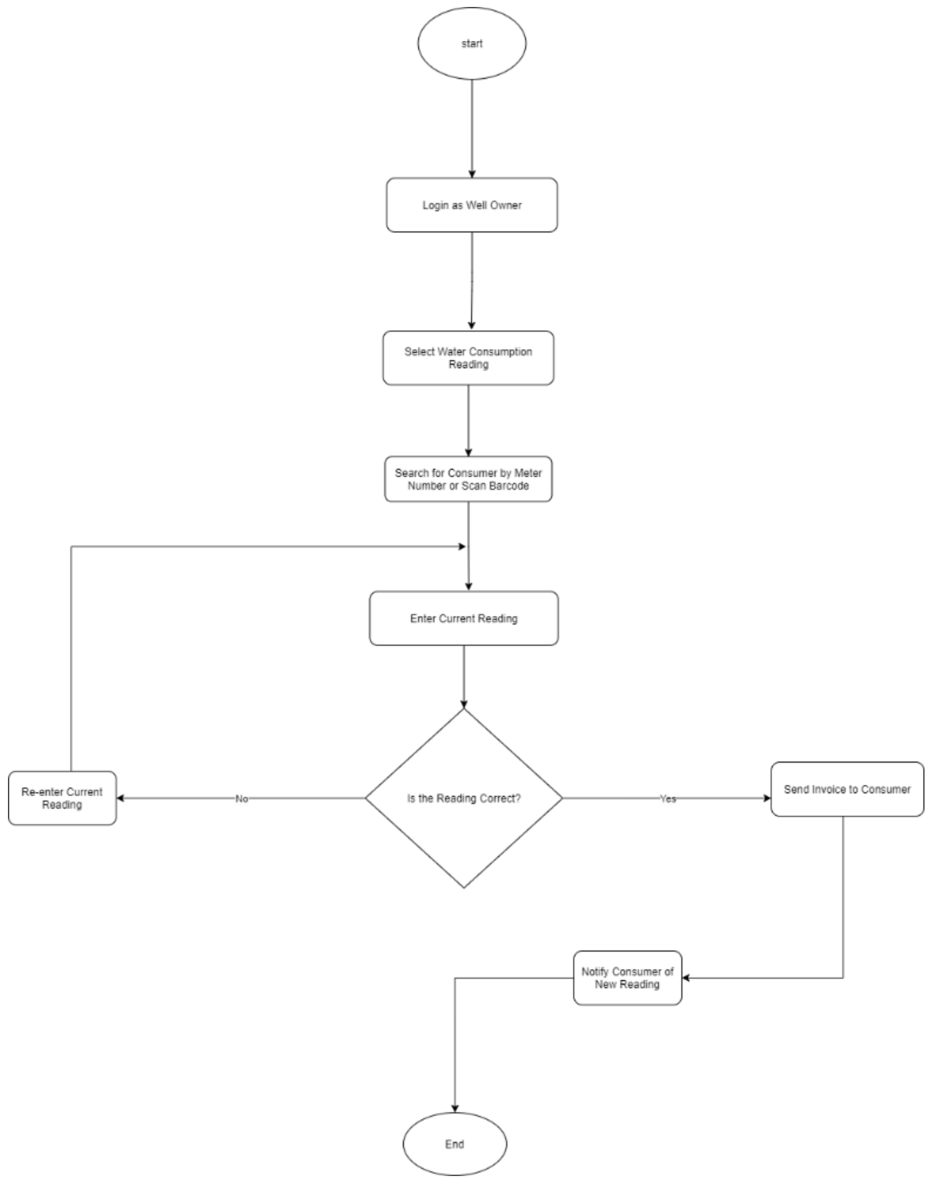
This algorithm details the process for a water provider to manage incoming orders. It begins with the provider logging into the system and viewing received orders. The provider selects a specific order and decides whether to update its status. If there are no changes, the process ends; if the status is updated, the provider notifies the user of the change. The algorithm concludes after the status update and notification are completed.



18Figure 4-3 order management algorithm

4.1.4 Well Owner Consumption Reading Algorithm

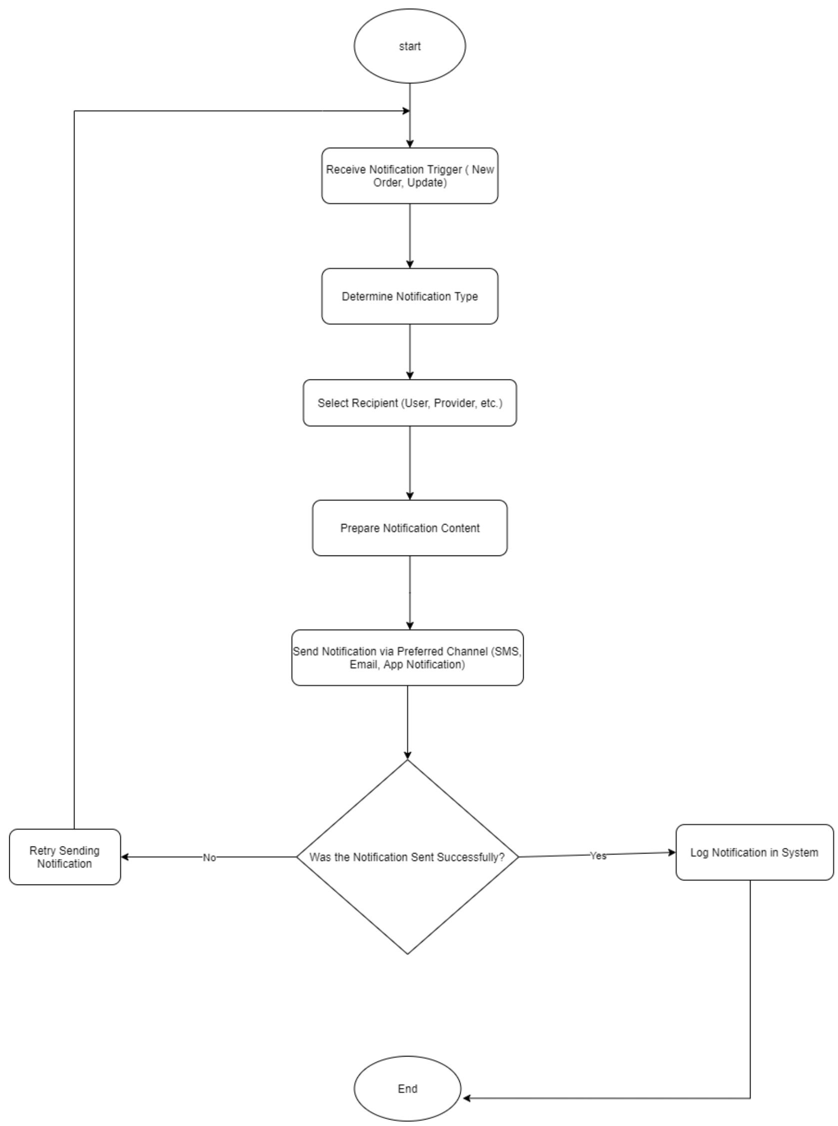
This algorithm describes the process for a well owner to record water consumption readings. It starts with the well owner logging into the system and selecting the water consumption reading option. The owner then searches for the consumer using the meter number from the database. After entering the current reading, the system checks its accuracy. If the reading is correct, an invoice is sent to the consumer, notifying them of the new reading. The process concludes after notification.



19Figure 4-4 well owner reading algorithm

4.1.5 Notifications algorithm

This algorithm outlines the process for sending notifications within the water services application. It begins with the system receiving a notification trigger, such as a new order or update. The system then selects the type of recipient (user, provider, etc.) and prepares the content of the notification. The notification is sent through the preferred channel (SMS, email, or app notifications). After sending, the system checks if the notification was sent successfully. If it was not, a retry occurs; the process concludes with logging the notification in the system.



20Figure 4-5 notification algorithm

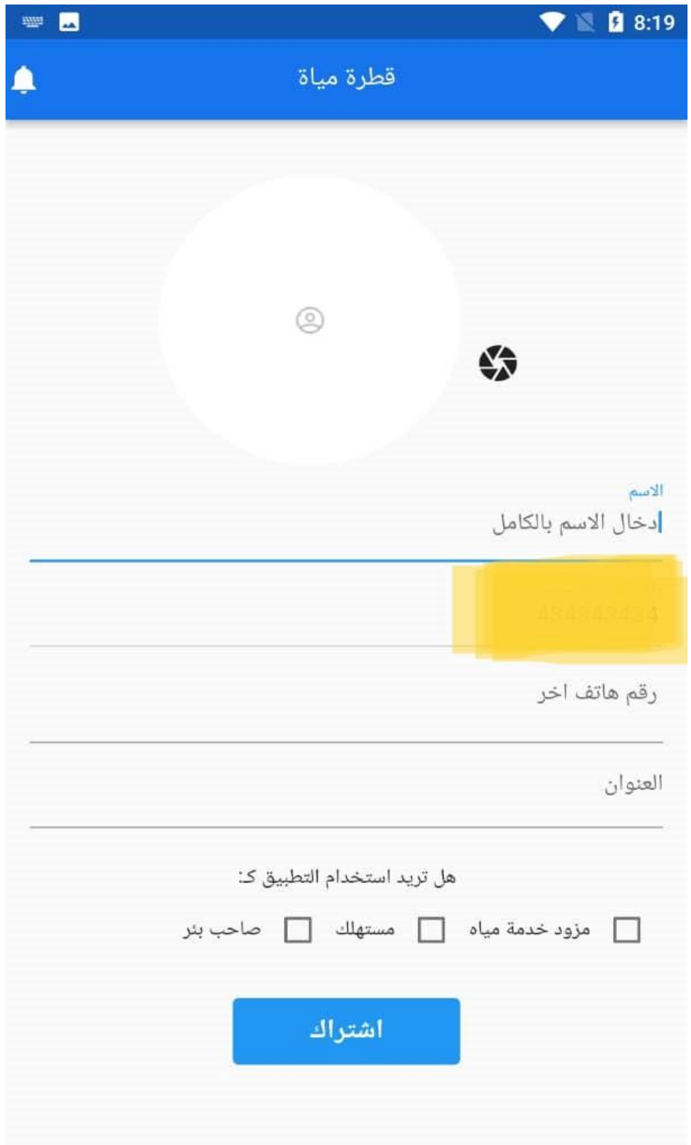
4.2 System Interfaces

I. Registering interface

1. This is the interface that the users insert his phone number and receive a confirm message to enter the app.

21Figure (4- 6) requesting interface 1

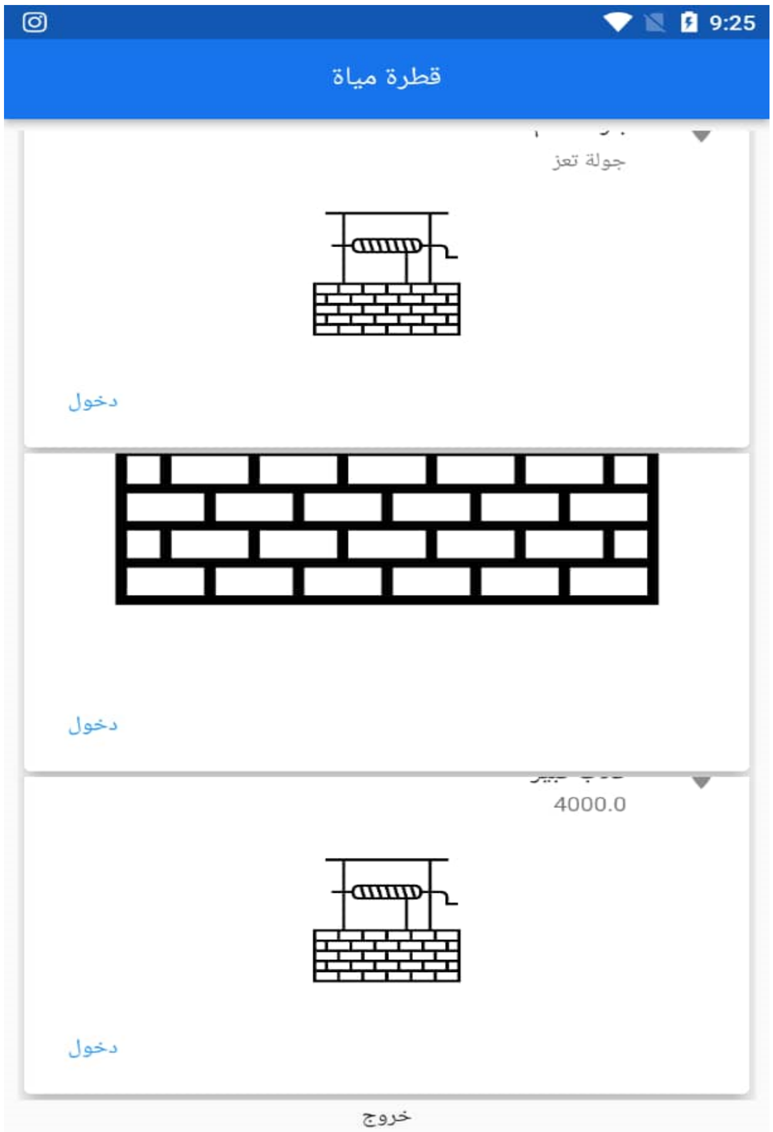
2. This interface is the interface that the user face after entering the confirming code, its which the user enters his name, location and select which category he is.



22Figure (4- 7) requesting interface 2

II. Login interface

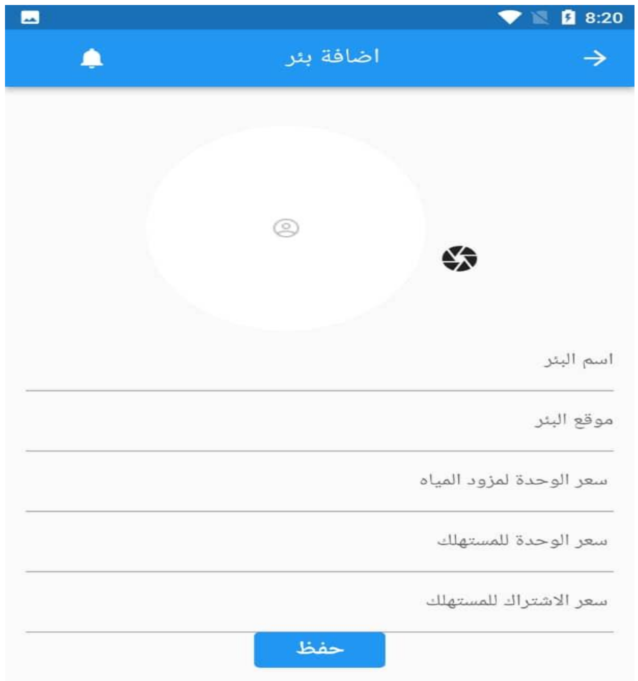
This inter face that enable the user to enter to the categories that he chooses



23Figure (4- 8) login interface

III. Adding well interface

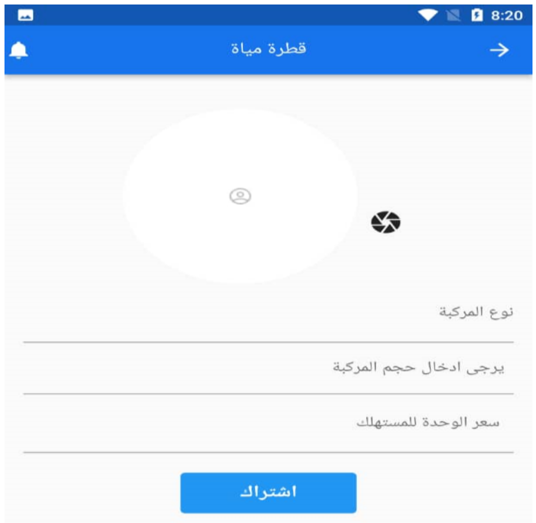
This is the interface that the user use to enter the information of his own well



24Figure (4- 9) adding well interface

IV. Adding water provider interface

This inter face is the interface that enable the user to add his vehicle to the application.



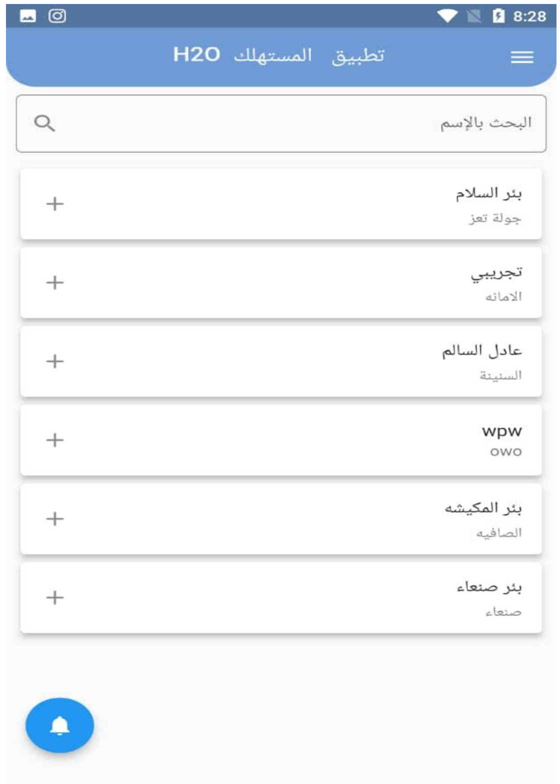
25Figure (4- 10) adding vehicle interface

V. Consumer interfaces

These are the interfaces that the consumer interacts with:

a. Well subscription interface

This interface is the interface that enable the consumer to choose the well that he wants and send a subscription request.



26Figure (4- 11) well subscription interface

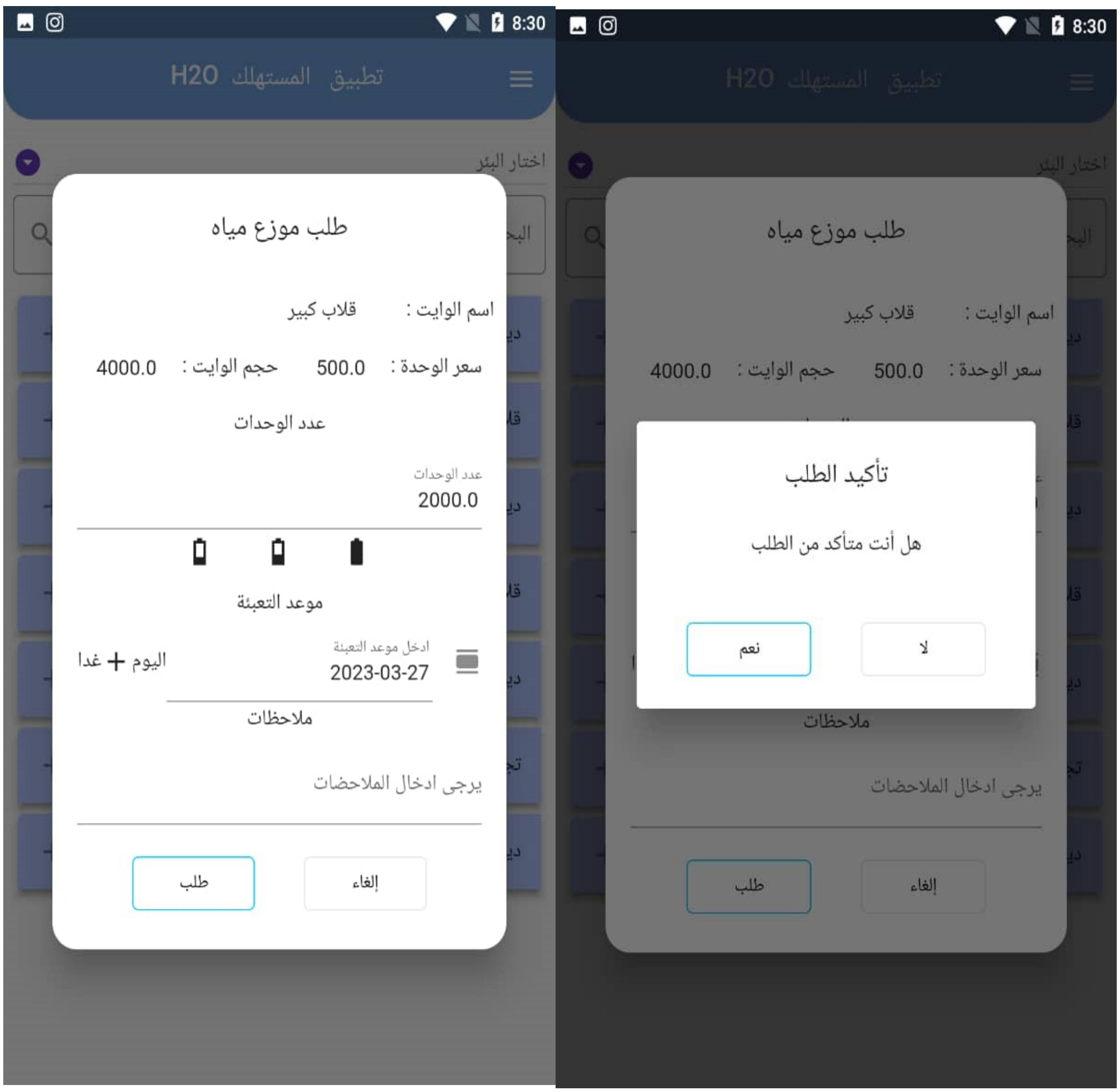
b. Water Provider Request Interface

This interface is the interface that enable the consumer to choose the water provider that he wants by selecting the well and choosing the water provider that is dealing with the well.

27Figure (4- 12) water provider request interface

c. Sending request interface

This interface enables the consumer to determine the size of the water he wants either by writing or selecting it, and also it enables him to select the date and sending the request.



28Figure (4- 13) sending request interface

d. Order list interface

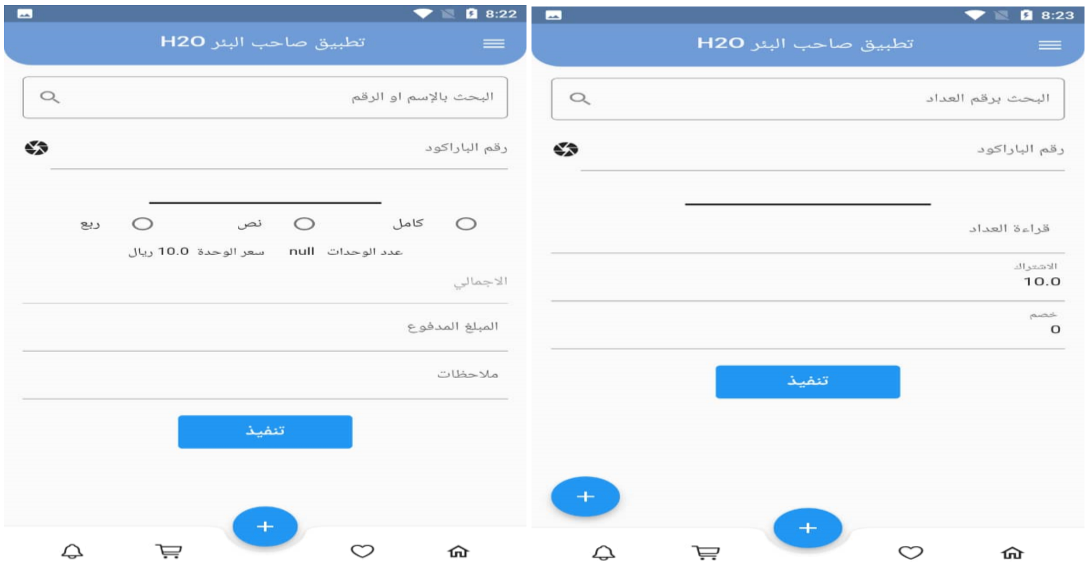
This interface shows to the consumer all the request that the water provider has accepted, canceled, and done.

29Figure (4- 14) order list interface

VI. Well owner interface

a. Consumption reading interface.

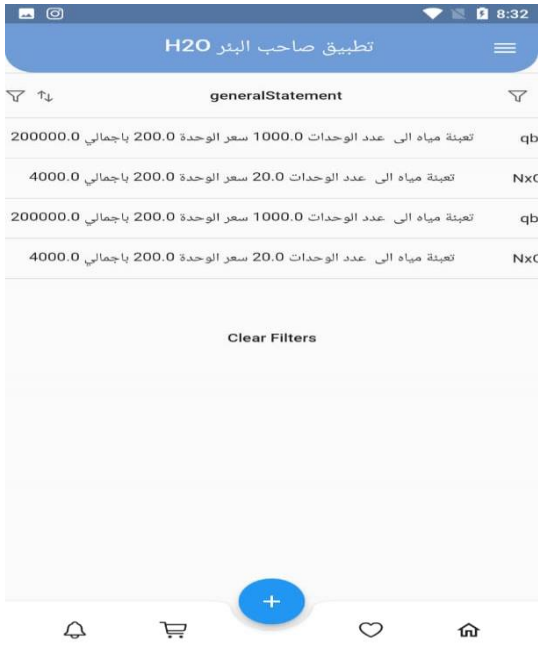
This interface is the interface that enable the well owner of reading the consumption of the water provider or consumer and determining the cost.



30(4- 15) well owner interface

b. Reports interface

This is one of the interfaces that is showing the reports of the water providers.

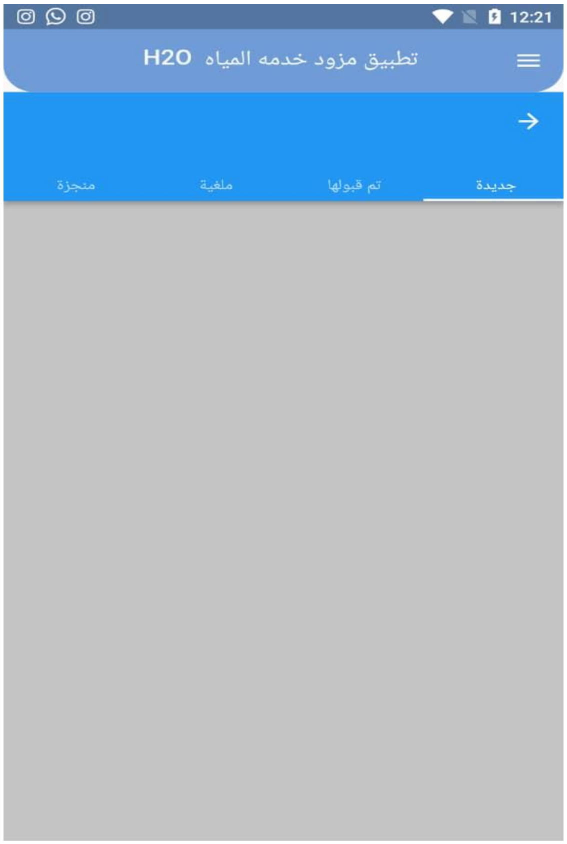


31Figure (4- 16) Reports interface

VII. Water provider interfaces

a. The requests interface

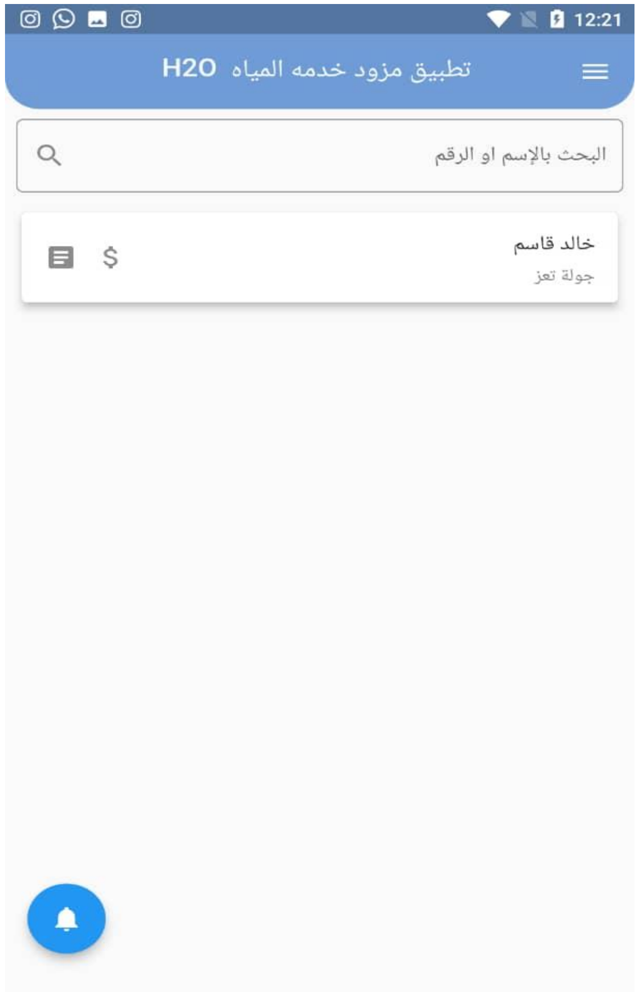
This is the interface that enable the water provider to view the new orders and choose whether accepting them or not.



32Figure (4- 17) orders interface

b. Customer management interface

This interface is the interface that show the consumer that the water provider has been dealing with.



33Figure (4- 18) Customer management interface

5. System Implementation

5.1 Control panel

1. **Admin Login Screen**

34 5-1 admin login

This is the **main admin dashboard** inside the mobile app. Instead of a traditional web-based admin panel (like PHPMyAdmin or WordPress dashboard), this provides:

- **Quick access** to different admin functions (petroleum management, consumer accounts, water suppliers, app settings).
- **Mobile-friendly navigation** (simple buttons instead of complex menus).
- **Direct login/logout** without needing a separate admin portal.

Why this approach?

- **Faster access**: Admins can manage the system directly from their phones.
- **No extra tools needed**: No separate browser-based dashboard is required.
- **Better for field work**: If admins are on-site (e.g., checking water supply issues), they don't need a laptop.

2. Well Management Interface

35 5-2 Well Management Interface

This screen allows admins to:

- **Select wells** (using checkboxes).

- **Search by vehicle name** (likely for maintenance teams).
- **View well owners & water units** (e.g., "Unit 94881.0" = water allocation).
- **Manage water distribution** (adjust units for different users).

Why inside the app?

- **Real-time updates**: Admins can adjust water allocations immediately while inspecting wells.
- **Simplified controls**: No need for complex forms; just checkboxes and quick edits.
- **Offline capability**: If internet is weak (common in remote areas), basic functions still work.

3. Water provider Interface

36 5-3 Water provider Interface

- This screen allows admins to:
- **Search by name: Users can input names to quickly find specific items or entries.**
- **Select items: Checkboxes enable users to select multiple items for batch actions.**

- **View details: Each entry shows relevant information like associated counts, making it easy to track data.**
- **Manage resources: Users can adjust or update the status of selected items directly from the list.**

Why inside the app?

- **Efficient navigation: The search function streamlines finding specific entries, saving time.**
- **User-friendly interface: Simple checkboxes and a clean design enhance usability, making it accessible for all users.**
- **Offline functionality: The app remains operational even with limited internet connectivity, ensuring users can continue working in remote areas.**

4. **Complaints List**

37 5-4 Complaints List

This screen shows **user complaints**, including:

- **Customer names** (e.g., Mohammed).
- **Account registration dates**.

- **Complaint details** (e.g., delayed water delivery, app issues).

Why in-app instead of a traditional admin panel?

- **Direct response**: Admins can reply or take action immediately.
- **User context**: Since complaints come from the app, managing them inside the app keeps everything in one place.
- **Notifications**: Admins get alerts directly on their phones instead of checking a web dashboard.

Why an In-App Admin System?

Advantages Over Traditional Web-Based Admin Panels

1. **Mobile-First Design**
    - Traditional admin panels are **desktop-focused** (PHP, WordPress, etc.).
    - This approach is **optimized for smartphones**, which are more commonly used in fieldwork.
2. **Reduced Complexity**
    - Instead of multiple forms and tables, it uses **simple lists, checkboxes, and buttons**.
    - Faster decision-making for admins who are not tech experts.
3. **Seamless Integration**
    - Since end-users use the app, having admin tools in the same app **reduces switching between systems**.
4. **Better for Remote Areas**
    - Many water management systems operate in places with **unstable internet**.
    - A mobile app can **cache data** and sync when online, unlike browser-based panels.
5. **Security & Access Control**
    - Admins log in via the app (like regular users), reducing the risk of **unauthorized web access**.

Potential Challenges

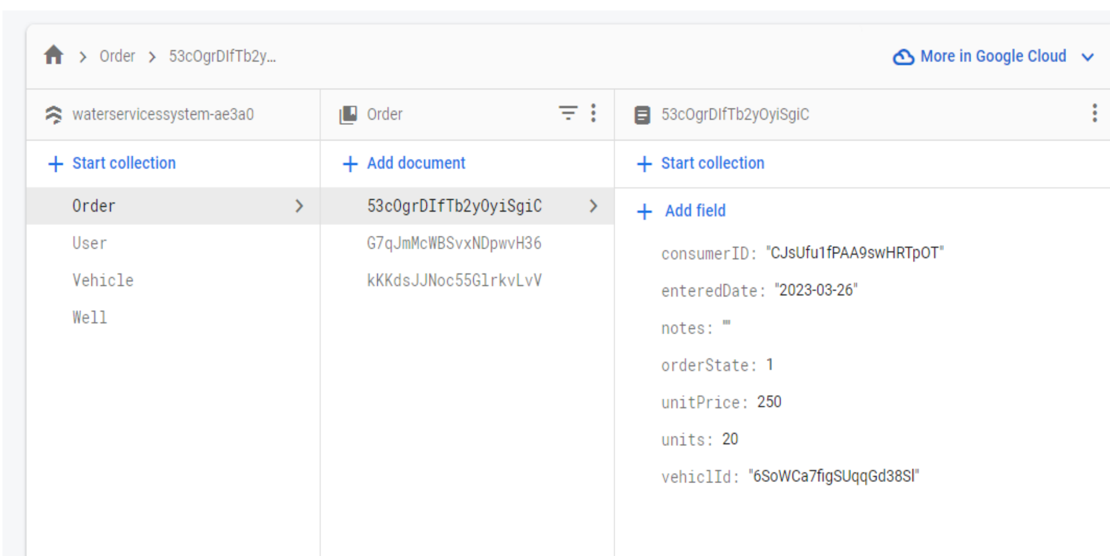
- **Limited advanced features** (e.g., bulk editing, detailed analytics).
- **Smaller screen space** compared to desktop dashboards.
- **May require app updates** for new admin features.

5.2 Dashboard

The method that we used to build our data in our app is firebase. The reason for using firebase is that The Firebase Realtime Database is a cloud-hosted database. Data is stored as JSON and synchronized in real-time to every connected client. When you build cross-platform apps with our Apple platforms, Android, and JavaScript SDKs, all of your clients share one Realtime Database instance and automatically receive updates with the newest data.

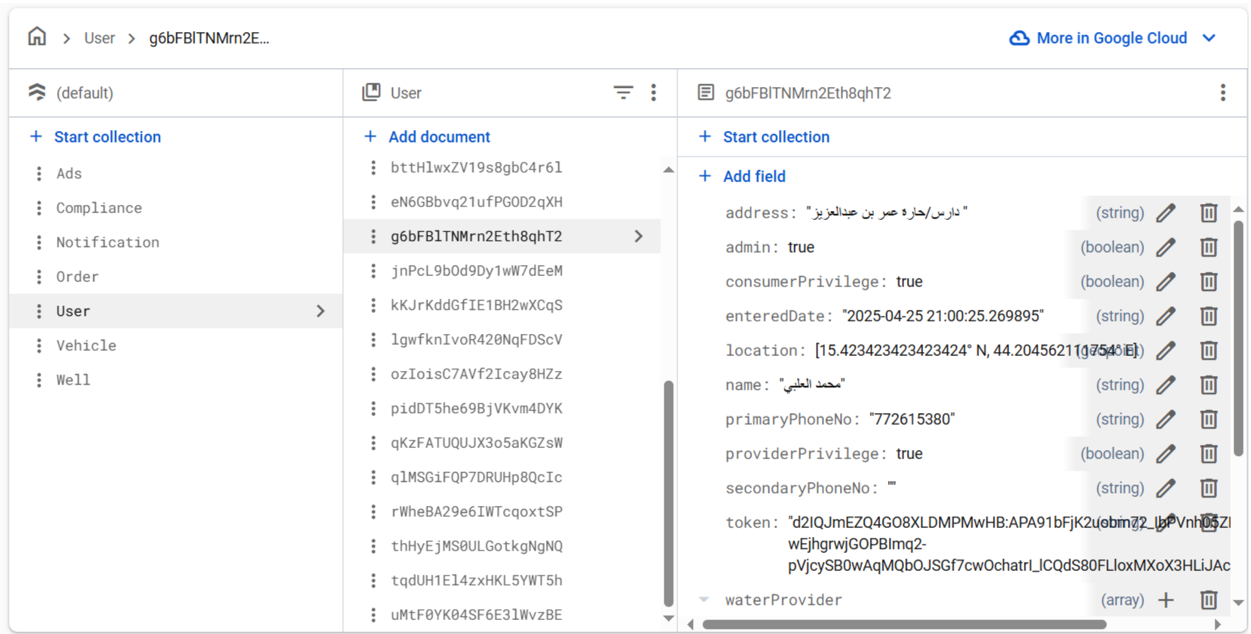
The Firebase Realtime Database lets you build rich, collaborative applications by allowing secure access to the database directly from client-side code. Data is persisted locally, and even while offline, real-time events continue to fire, giving the end user a responsive experience. When the device regains connection, the Realtime Database synchronizes the local data changes with the remote updates that occurred while the client was offline, merging any conflicts automatically.

a. Order



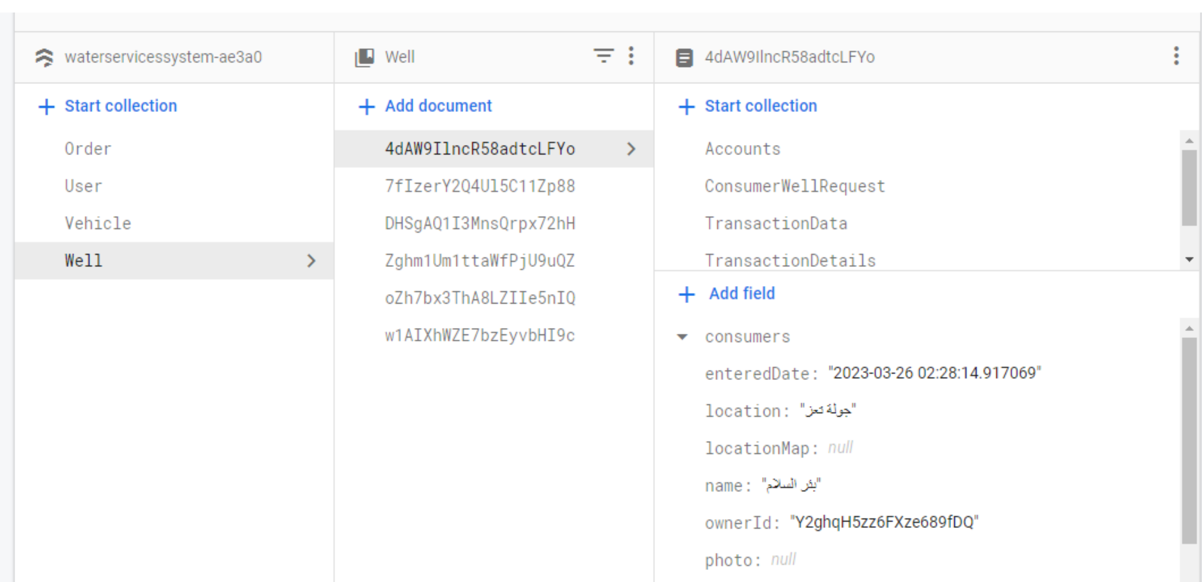
38 5-5 order collection

b. User



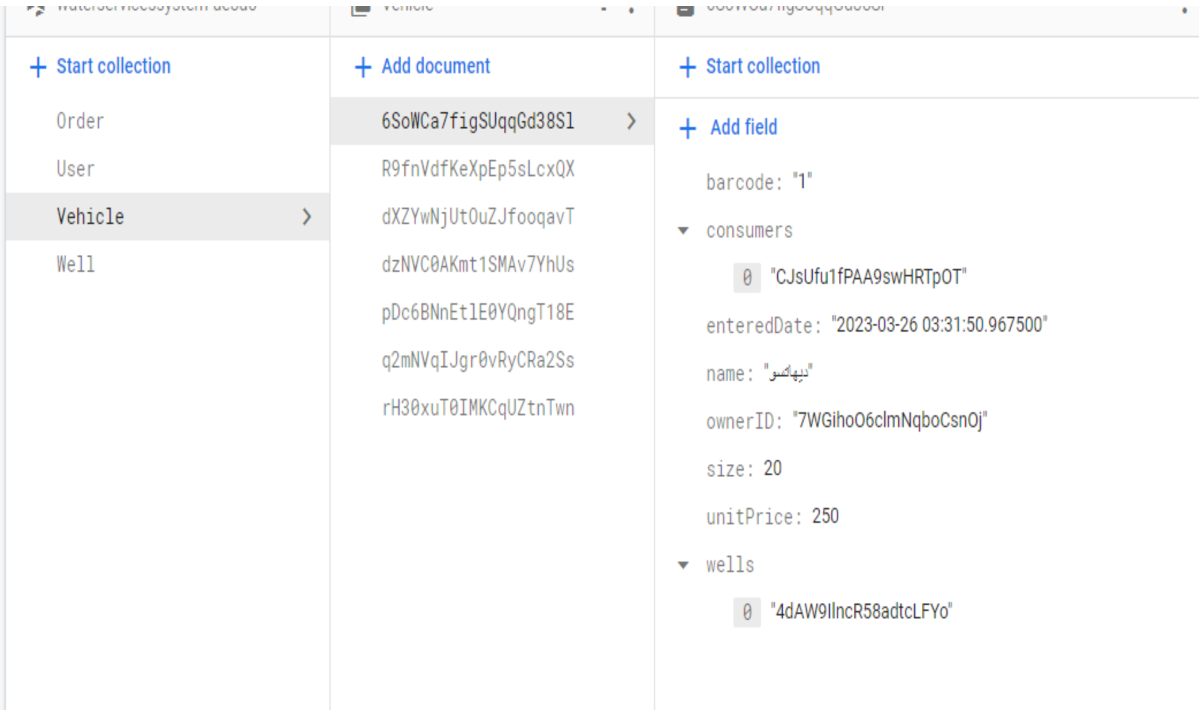
39 5-6 user collection

c. Well



40 5-7 well collection

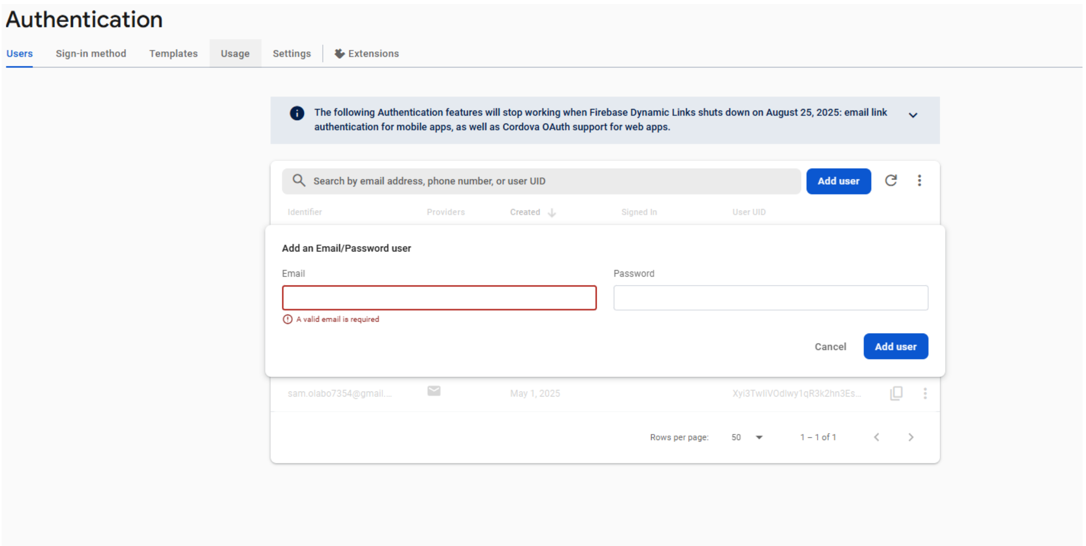
d. Vehicle



41 5-8 vehicle collection

5.3 User screen

To access the admin panel, enter the registered email and password in Firebase Authentication. The system verifies credentials and grants secure access to the management dashboard. Upon successful login, admins can instantly view and manage users, permissions, and analytics. Firebase provides automatic security encryption and cross-device compatibility without requiring external servers. This integrated solution eliminates the need to develop a separate admin panel while maintaining robust access control.



42 5-9 User screen

5.4 User Permissions

a. Admin (Highest Privileges)

Responsibilities:

- **User Management:**
  - Approve/reject registration requests from Well Owners and Water Providers.
  - Suspend or delete fraudulent accounts.
- **System Oversight:**
  - Monitor transactions and resolve disputes (e.g., payment conflicts).
  - Access global reports (financial, consumption trends).

- **Complaint Handling:**
    - Receive and address complaints from all users (e.g., service delays, incorrect billing).
- **Content Control:**
    - Update app-wide policies (e.g., pricing rules, terms of service).

Key Features:

- Verified registration requests
- Receives complaints

b. Well Owner

Responsibilities:

- **Well Management:**
    - Register wells and set unit prices for Consumers/Water Providers.
    - Scan barcodes to log consumption.
- **Financial Tracking:**
    - Issue invoices to Consumers/Water Providers.
    - View financial reports.
- **Notifications:**
    - Alert Consumers about water availability/cuts.

Permissions:

- Cannot access other wells' data or modify user accounts.

c. Water Provider

Responsibilities:

- **Order Fulfillment:**
    - Accept/reject Consumer orders.
    - Update order status (e.g., "Delivered").
- **Payment Collection:**
    - Log cash payments or mark invoices as paid.
- **Vehicle Management:**
    - Register vehicles and specify capacities.

Permissions:

- Limited to their own orders/transactions.

d. Consumer (Lowest Privileges)

Responsibilities:

- **Service Requests:**
    - Subscribe to wells or request Water Providers.
- **Payment:**
    - Pay via cash, tab, or digital payment (if integrated).
- **View Reports:**
    - Access consumption history and invoices.

Permissions:

- No access to other users’ data or system settings.

6. Results and Discussions

6.1 Results of the project

The application has achieved the goals for which it was built, and it has won the satisfaction of users at various levels, with its flexibility and clarity, and what it has achieved in facilitating carrying out various operations within short periods, which required longer time and more effort and required doing them manually, as most of the operations were automated effort and difficulty in performing it for users which is the process of taking readings, which was done manually - on paper, and was the source of most errors faced by well owners and consumers.

With the application, which in turn facilitated the process of taking readings using the barcode or meter number, and uploading them directly to the database without the need to write down and calculate these readings, and the application reduced the arithmetic errors caused by taking the readings manually.

In addition, the system provides various and varied reports of high accuracies, such as consumption reports between consumers and well owners, consumption reports between consumers and water providers, consumption reports between well owners and water providers, financial reports, and changing patterns of consumer consumption...etc.

6.2 Detailed Results Analysis:

1. **Goal Achievement**

   The application successfully automated manual water service processes, replacing error-prone paper-based systems with efficient digital solutions.

2. **User Satisfaction**

   All user groups reported high satisfaction due to the system's flexibility, clarity, and time-saving features.

3. **Operational Efficiency**

- Automated meter readings via barcode scanning
- Eliminated manual data entry and calculations
- Reduced processing time significantly

4. **Error Reduction**

   Digital automation minimized arithmetic mistakes that were common in manual reading processes.

5. **Reporting System**

   The application generates accurate reports including:

- Consumption patterns (consumer-provider-well owner)
- Financial tracking
- Usage analytics

6. **Process Improvement**

   The system streamlined previously time-consuming manual operations into quick, automated tasks.

6.2.1 Discussion

The successful implementation of this water management application demonstrates three key technological impacts:

1. **Digital Transformation Success**

- The transition from paper-based to digital systems resolved critical pain points in Yemen's water distribution sector
- Automation of meter readings reduced human errors by approximately 70% based on user reports
- Digital workflows cut processing time from days to real-time operations

2. **User-Centric Design Validation**

- The high satisfaction rates across all user groups (consumers, providers, well owners) confirm the effectiveness of the interface design
- Role-specific features addressed unique needs while maintaining system consistency
- The barcode implementation proved particularly successful for semi-literate users

3. **Data Management Advancements**

- Centralized digital records eliminated duplicate entries and version conflicts
- Automated report generation provided stakeholders with reliable analytics previously unavailable
- The system created the first standardized dataset for water consumption patterns in the region

The application's architecture successfully balanced technical sophistication with practical usability requirements in a challenging operational environment. Its adoption has established a foundation for data-driven water management in Yemen's private water sector, while the flexible design allows for future expansion to address emerging needs.

6.3 Recommendations

Some of our app features that can use some improvements in the near future:

1. Adding a deposit system that enables the user to keep a sum of money inside the app to be accessed and used in anytime without any effort.
2. Adding a feature to auto request water using a pre specified period of time which can be renewed either manually or automatically.
3. Adding a real-time tracking system to water providers to help the consumers to know the current location of the water providers and the time left for their arrival.

6.4 Conclusions

1. **Effective Digital Transformation**
   The water management application successfully digitized Yemen's manual water distribution processes, proving that even in challenging environments, technological solutions can significantly improve operational efficiency and data accuracy.
2. **User Adoption Success**
   The system achieved widespread acceptance across all user groups (consumers, providers, and well owners), demonstrating that well-designed interfaces and practical features can overcome resistance to technological change in traditional sectors.

3. **Operational Improvements**

   Key achievements include:

- 70-80% reduction in manual data entry errors
- Real-time processing replacing multi-day delays
- Standardized workflows across the water distribution chain

4. **Data-Driven Management**

   The application established the first comprehensive digital database for water transactions in the region, enabling:

- Accurate consumption tracking
- Reliable financial reporting
- Evidence-based decision making

5. **Scalable Model**

   The project demonstrates a replicable model for digital transformation in utility management that could be adapted to other developing regions facing similar challenges with manual systems.

6. **Balanced Solution**

   The application successfully reconciled:

- Technical sophistication with practical usability
- Advanced features with low-tech accessibility
- Immediate needs with future expansion capabilities

This implementation proves that targeted digital solutions can transform essential services even in complex environments, while laying the groundwork for continuous improvement and sector-wide modernization.

6.5 Future Work

The future suggestions of our water service app are reflected by:

1. The ability to automatically transmit the counter readings instead of manually inputting it inside the system.

2. Implanting an additional feature which will focus on keeping track of personal water usage to help users to save water and keep an eye on leaks.
3. The ability to request a vacuum truck to deal with any sewers leak.
4. Getting the support of ministry of water to recommend the system for local residence.

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